Conditional Memory via Scalable Lookup: A New Axis of Sparsity for Large Language Models

Xin Cheng[1,2], Wangding Zeng[2], Damai Dai[2], Qinyu Chen[2], Bingxuan Wang[2], Zhenda Xie[2], Kezhao Huang[2], Xingkai Yu[2], Zhewen Hao[2], Yukun Li[2], Han Zhang[2], Huishuai Zhang[1], Dongyan Zhao[1], Wenfeng Liang[2]

[1]Peking University [2]DeepSeek-AI

{zhanghuishuai, zhaody}@pku.edu.cn

{chengxin, zengwangding, damai.dai}@deepseek.com

Abstract

While Mixture-of-Experts (MoE) scales capacity via conditional computation, Transformers lack a native primitive for knowledge lookup, forcing them to inefficiently simulate retrieval through computation. To address this, we introduce conditional memory as a complementary sparsity axis, instantiated via **Engram**, a module that modernizes classic N-gram embedding for $O(1)$ lookup. By formulating the *Sparsity Allocation* problem, we uncover a U-shaped scaling law that optimizes the trade-off between neural computation (MoE) and static memory (Engram). Guided by this law, we scale Engram to 27B parameters, achieving superior performance over a strictly iso-parameter and iso-FLOPs MoE baseline. Most notably, while the memory module is expected to aid knowledge retrieval (e.g., MMLU +3.4; CMMLU +4.0), we observe even larger gains in general reasoning (e.g., BBH +5.0; ARC-Challenge +3.7) and code/math domains (HumanEval +3.0; MATH +2.4). Mechanistic analyses reveal that Engram relieves the backbone's early layers from static reconstruction, effectively deepening the network for complex reasoning. Furthermore, by delegating local dependencies to lookups, it frees up attention capacity for global context, substantially boosting long-context retrieval (e.g., Multi-Query NIAH: 84.2 → 97.0). Finally, Engram establishes infrastructure-aware efficiency: its deterministic addressing enables runtime prefetching from host memory, incurring negligible overhead. We envision conditional memory as an indispensable modeling primitive for next-generation sparse models. Code available at: `https://github.com/deepseek-ai/Engram`

1. Introduction

Sparsity is a recurring design principle for intelligent systems, spanning from biological neural circuits (Lennie, 2003; Olshausen and Field, 1997) to modern Large Language Models (LLMs). Currently, this principle is primarily realized through Mixture-of-Experts (MoE) (Dai et al., 2024; Shazeer et al., 2017), which scales capacity via conditional computation. Owing to its ability to drastically increase model size without proportional increases in compute, MoE has become the de facto standard for frontier models (Comanici et al., 2025; Guo et al., 2025; Team et al., 2025).

Despite the success of this conditional computation paradigm, the intrinsic heterogeneity of linguistic signals suggests significant room for *structural optimization*. Specifically, language modeling entails two qualitatively different sub-tasks: compositional reasoning and knowl-

1 # 通过可扩展查找实现的条件记忆：
2 大语言模型稀疏性的新维度

程欣 [1,2], 曾望鼎 [2], 戴大迈 [2], 陈沁宇 [2], 王炳轩 [2],
谢振达 [2], 黄科钊 [2], 余星凯 [2], 郝哲文 [2], 李宇坤 [2], 张晗 [2],
3 张辉帅 [1], 赵东岩 [1], 梁文锋 [2]

4 [1] 北京大学 [2] 深度求索人工智能公司

5 {zhanghuishuai, zhaody}@pku.edu.cn

{chengxin, zengwangding, damai.dai}@deepseek.com

6 ## Abstract

7 虽然专家混合模型（MoE）通过条件计算扩展了模型容量，但 Transformer 缺乏原生的知识查找原语，迫使其通过计算低效地模拟检索过程。为解决此问题，我们引入了条件记忆作为补充的稀疏化维度，并通过 **Engram** 模块实现，该模块将经典的 N-gram 嵌入现代化，用于 $O(1)$ 查找。通过形式化 *Sparsity Allocation* 问题，我们发现了一个 U 形缩放定律，该定律优化了神经计算（MoE）与静态记忆（Engram）之间的权衡。在此定律指导下，我们将 Engram 扩展到 270 亿参数，其性能优于严格的等参数和等 FLOPs 的 MoE 基线。最值得注意的是，虽然预期记忆模块有助于知识检索（例如 MMLU +3.4；CMMLU +4.0），但我们观察到在通用推理（例如 BBH +5.0；ARC-Challenge +3.7）和代码/数学领域（HumanEval +3.0；MATH +2.4）有更大的增益。机制分析表明，Engram 解除了主干网络早期层进行静态重构的负担，有效地加深了网络以进行复杂推理。此外，通过将局部依赖关系委托给查找操作，它释放了注意力机制处理全局上下文的能力，从而显著提升了长上下文检索性能（例如 Multi-Query NIAH：84.2 → 97.0）。最后，Engram 实现了面向基础设施的效率：其确定性寻址机制支持从主机内存进行运行时预取，产生的开销可忽略不计。我们设想条件记忆将成为下一代稀疏模型中不可或缺的建模原语。代
8 码发布于：`https://github.com/deepseek-ai/Engram`

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edge retrieval. While the former demands deep, dynamic computation, a substantial portion of text—such as named entities and formulaic patterns—is local, static, and highly stereotyped (Constant et al., 2017; Erman, 2000). The effectiveness of classical N-gram models (Brants et al., 2007; Liu et al., 2024b; Nguyen, 2024) in capturing such local dependencies implies that these regularities are naturally represented as computationally inexpensive lookups. Since standard Transformers (Vaswani et al., 2017) lack a native knowledge lookup primitive, current LLMs are forced to **simulate retrieval through computation**. For instance, resolving a common multi-token entity requires consuming multiple early layers of attention and feed-forward networks (Ghandeharioun et al., 2024; Jin et al., 2025) (see Table 3). This process essentially amounts to an expensive runtime reconstruction of a static lookup table, wasting valuable sequential depth on trivial operations that could otherwise be allocated to higher-level reasoning.

To align model architecture with this linguistic duality, we advocate for a complementary axis of sparsity: conditional memory. Whereas conditional computation sparsely activates parameters to process dynamic logic (Bengio et al., 2013; Shazeer et al., 2017), conditional memory relies on sparse lookup operations to retrieve static embeddings for fixed knowledge. As a preliminary exploration of this paradigm, we revisit N-gram embeddings (Bojanowski et al., 2017) as a canonical instantiation: local context serves as a key to index a massive embedding table via constant-time $O(1)$ lookups (Huang et al., 2025a; Pagnoni et al., 2025; Tito Svenstrup et al., 2017; Yu et al., 2025). Our investigation reveals that, perhaps surprisingly, this static retrieval mechanism can serve as an ideal complement to modern MoE architecture—but only if it is properly designed. In this paper, we propose **Engram**, a conditional memory module grounded in the classic N-gram structure but equipped with modern adaptations such as tokenizer compression, multi-head hashing, contextualized gating, and multi-branch integration (detailed in Section 2).

To quantify the synergy between these two primitives, we formulate the *Sparsity Allocation* problem: given a fixed total parameter budget, how should capacity be distributed between MoE experts and Engram memory? Our experiments uncover a distinct U-shaped scaling law, revealing that even simple lookup mechanisms, when treated as a first-class modeling primitive, act as essential complements to neural computation. Guided by this allocation law, we scale Engram to a 27B-parameter model. Compared to a strictly iso-parameter and iso-FLOPs MoE baseline, Engram-27B achieves superior efficiency across diverse domains. Crucially, the gains are not limited to knowledge-intensive tasks (e.g., MMLU: +3.4; CMMLU: +4.0; MMLU-Pro: +1.8), where memory capacity is intuitively beneficial; we observe even more significant improvements in general reasoning (e.g., BBH: +5.0; ARC-Challenge: +3.7; DROP: +3.3) and code/math domains (e.g., HumanEval: +3.0; MATH: +2.4; GSM8K: +2.2).

Mechanistic analysis via LogitLens (nostalgebraist, 2020) and CKA (Hendrycks et al., 2021a) reveals the source of these gains: Engram relieves the backbone from reconstructing static knowledge in early layers, thereby increasing effective depth available for complex reasoning. Furthermore, by delegating local dependencies to lookups, Engram frees up attention capacity to focus on global context, enabling exceptional performance in long-context scenarios—substantially outperforming baselines on LongPPL (Fang et al.) and RULER (Hsieh et al.) (e.g., Multi-Query NIAH: 97.0 vs. 84.2; Variable Tracking: 89.0 vs. 77.0).

Finally, we establish infrastructure-aware efficiency as a first-class principle. Unlike MoE's dynamic routing, Engram employs deterministic IDs to enable runtime prefetching, overlapping communication with computation. Empirical results show that offloading a 100B-parameter table to host memory incurs negligible overhead (< 3%). This demonstrates that Engram effectively bypasses GPU memory constraints, facilitating aggressive parameter expansion.

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1. Introduction

稀疏性是智能系统的一个反复出现的设计原则，从生物神经回路 (Lennie, 2003; Olshausen and Field, 1997) 到现代大语言模型（LLMs）皆是如此。目前，这一原则主要通过混合专家（MoE）

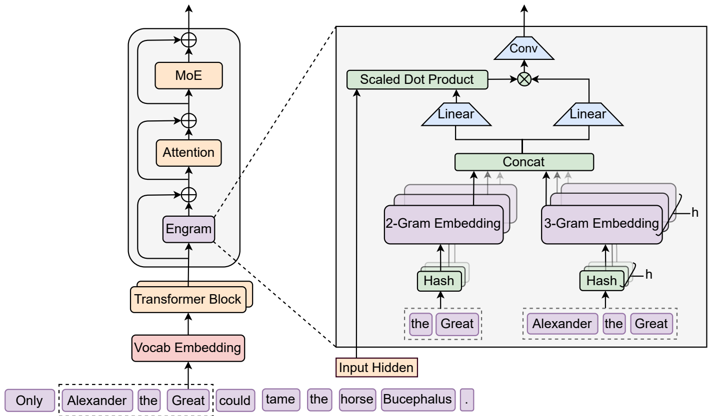


Figure 1 | **The Engram Architecture.** The module augments the backbone by retrieving static N-gram memory and fusing it with dynamic hidden states via context-aware gating. This module is applied only to specific layers to decouple memory from compute, leaving the standard input embedding and un-embedding module intact.

2. Architecture

2.1. Overview

As shown in Figure 1, Engram is a conditional memory module designed to augment the Transformer backbone by structurally separating static pattern storage from dynamic computation. Formally, given an input sequence $X = (x_1, \ldots, x_T)$ and hidden states $\mathbf{H}^{(\ell)} \in \mathbb{R}^{T \times d}$ at layer ℓ, the module processes each position t in two functional phases: **retrieval and fusion**. First, as detailed in Section 2.2, we extract and compress suffix N-grams to deterministically retrieve static embedding vectors via hashing. Subsequently, in Section 2.3, these retrieved embeddings are dynamically modulated by the current hidden state and refined via a lightweight convolution. Finally, we discuss the integration with multi-branch architectures in Section 2.4 and the system-level design in Section 2.5.

2.2. Sparse Retrieval via Hashed N-grams

The first phase maps local contexts to static memory entries, involving tokenizer compression and retrieving embeddings via deterministic hashing.

Tokenizer Compression While N-gram models typically operate directly on tokenizer outputs, standard subword tokenizers prioritize *lossless reconstruction*, often assigning disjoint IDs to semantically equivalent terms (e.g., `Apple` vs. `␣apple`) (Kudo and Richardson, 2018; Li et al., 2023b). To maximize semantic density, we implement a vocabulary projection layer. Specifically, we pre-compute a surjective function $\mathcal{P} : V \to V'$ that collapses raw token IDs into canonical

(Dai et al., 2024; Shazeer et al., 2017) 实现，它通过条件计算来扩展模型容量。得益于其能够大幅增加模型规模而无需按比例增加计算量，MoE 已成为前沿模型 (Comanici et al., 2025; Guo et al., 2025; Team et al., 2025) 的事实标准。

尽管这种条件计算范式取得了成功,但语言信号的内在异质性表明在 *structural optimization* 方面仍有巨大改进空间。具体而言，语言建模包含两个性质不同的子任务：组合推理和知识检索。前者需要深度、动态的计算，而文本的很大一部分——例如命名实体和公式化模式——是局部的、静态的且高度刻板的 (Constant et al., 2017; Erman, 2000)。经典的 N-gram 模型 (Brants et al., 2007; Liu et al., 2024b; Nguyen, 2024) 在捕捉此类局部依赖关系方面的有效性意味着，这些规律性很自然地表现为计算成本低廉的查找操作。由于标准 Transformer (Vaswani et al., 2017) 缺乏原生的知识查找原语，当前的大语言模型被迫 **simulate retrieval through computation**。例如，解析一个常见的多词元实体需要消耗注意力机制和前馈网络 (Ghandeharioun et al., 2024; Jin et al., 2025) 的多个早期层（参见表 3）。这个过程本质上相当于对一个静态查找表进行昂贵的运行时重建，将宝贵的序列深度浪费在琐碎的操作上，而这些深度本可以分配给更高层次的推理。

为了使模型架构与这种语言二元性保持一致，我们主张一个互补的稀疏性维度：条件记忆。条件计算稀疏地激活参数以处理动态逻辑 (Bengio et al., 2013; Shazeer et al., 2017)，而条件记忆则依赖于稀疏的查找操作来检索固定知识的静态嵌入。作为对该范式的初步探索，我们重新审视 N-gram 嵌入 (Bojanowski et al., 2017) 作为一种典型实例：局部上下文作为键，通过恒定时间的 $O(1)$ 查找 (Huang et al., 2025a; Pagnoni et al., 2025; Tito Svenstrup et al., 2017; Yu et al., 2025) 来索引一个庞大的嵌入表。我们的研究表明，或许令人惊讶的是，这种静态检索机制可以成为现代 MoE 架构的理想补充——但前提是设计得当。在本文中，我们提出 **Engram**，这是一个基于经典 N-gram 结构的条件记忆模块，但配备了现代化的改进，例如分词器压缩、多头哈希、上下文门控和多分支集成（详见节 2）。为了量化这两种基本组件之间的协同效应，我们提出了 *Sparsity Allocation* 问题：在固定的总参数量预算下，应如何在 MoE 专家和 Engram 记忆之间分配容量？我们的实验揭示了一个独特的 U 形缩放定律，表明即使是简单的查找机制，当被视为一等建模基元时，也能作为神经计算的重要补充。

依据此分配定律的指导，我们将 Engram 扩展为一个 270 亿参数的模型。与严格同参数、同 FLOPs 的 MoE 基线相比，Engram-27B 在多个领域实现了更优的效率。关键的是，其优势不仅限于知识密集型任务（例如，MMLU：+3.4；CMMLU：+4.0；MMLU-Pro：+1.8），在这些任务中记忆容量直观上是有益的；我们甚至在通用推理（例如，BBH：+5.0；ARC-Challenge：+3.7；DROP：+3.3）以及代码/数学领域（例如，HumanEval：+3.0；MATH：+2.4；GSM8K：+2.2）观察到了更显著的改进。

通过 LogitLens (nostalgebraist, 2020) 和 CKA (Hendrycks et al., 2021a) 进行的机制分析揭示了这些收益的来源：Engram 减轻了主干网络在早期层重构静态知识的负担，从而增加了可用于复杂推理的有效深度。此外，通过将局部依赖关系委托给查找操作，Engram 释放了注意力容量以专注于全局上下文，从而在长上下文场景中实现了卓越的性能——在 LongPPL (Fang et al.)

identifiers based on normalized textual equivalence (using NFKC (Whistler, 2025), lowercasing, etc.). In practice, this process achieves a 23% reduction in the effective vocabulary size for a 128k tokenizer (see Appendix C). Formally, for a token at position t, we map its raw ID x_t to a canonical ID $x'_t = \mathcal{P}(x_t)$ to form the suffix N-gram $g_{t,n} = (x'_{t-n+1}, \ldots, x'_t)$.

Multi-Head Hashing. Directly parameterizing the combinatorial space of all possible N-grams is intractable. Following Tito Svenstrup et al. (2017), we adopt a hashing-based approach. To mitigate collisions, we employ K distinct hash heads for each N-gram order n. Each head k maps the compressed context to an index within an embedding table $\mathbf{E}_{n,k}$ (of prime size $M_{n,k}$) via a deterministic function $\varphi_{n,k}$:

$$z_{t,n,k} \triangleq \varphi_{n,k}(g_{t,n}), \quad \mathbf{e}_{t,n,k} = \mathbf{E}_{n,k}[z_{t,n,k}]. \tag{1}$$

In practice, $\varphi_{n,k}$ is implemented as a lightweight multiplicative-XOR hash. We construct the final memory vector $\mathbf{e}_t \in \mathbb{R}^{d_{\text{mem}}}$ by concatenating all retrieved embeddings:

$$\mathbf{e}_t \triangleq \mathop{\Big\|}_{n=2}^{N} \mathop{\Big\|}_{k=1}^{K} \mathbf{e}_{t,n,k}. \tag{2}$$

2.3. Context-aware Gating

The retrieved embeddings $\mathbf{e}_t$ serve as context-independent priors. Being static, however, they inherently lack contextual adaptability and may suffer from noise due to hash collisions or polysemy (Haber and Poesio, 2024). To enhance expressivity and resolve this ambiguity, we employ a context-aware gating mechanism inspired by Attention (Bahdanau et al., 2015; Vaswani et al., 2017). Specifically, we utilize the current hidden state $\mathbf{h}_t$—which has aggregated global context via preceding attention layers—as a dynamic Query, while the retrieved memory $\mathbf{e}_t$ serves as the source for both Key and Value projections:

$$\mathbf{k}_t = \mathbf{W}_K \mathbf{e}_t, \quad \mathbf{v}_t = \mathbf{W}_V \mathbf{e}_t \tag{3}$$

where $\mathbf{W}_K, \mathbf{W}_V$ are learnable projection matrices. To ensure gradient stability (Dehghani et al., 2023), we apply RMSNorm (Zhang and Sennrich, 2019) to the Query and Key before computing the scalar gate $\alpha_t \in (0, 1)$:

$$\alpha_t = \sigma\left(\frac{\text{RMSNorm}(\mathbf{h}_t)^\top \text{RMSNorm}(\mathbf{k}_t)}{\sqrt{d}}\right). \tag{4}$$

The gated output is defined as $\tilde{\mathbf{v}}_t = \alpha_t \cdot \mathbf{v}_t$. This design enforces semantic alignment: if the retrieved memory $\mathbf{e}_t$ contradicts the current context $\mathbf{h}_t$, the gate α_t tends toward zero, effectively suppressing the noise.

Finally, to expand the receptive field and enhance the model's non-linearity, we introduce a short, depthwise causal convolution (Gu et al., 2022; Peng et al., 2023). Let $\tilde{\mathbf{V}} \in \mathbb{R}^{T \times d}$ denote the sequence of gated values. Using a kernel size w (set to 4), dilation δ (set to the max N-gram order) and SiLU activation (Elfwing et al., 2018), the final output $\mathbf{Y}$ is computed as:

$$\mathbf{Y} = \text{SiLU}\left(\text{Conv1D}(\text{RMSNorm}(\tilde{\mathbf{V}}))\right) + \tilde{\mathbf{V}}, \tag{5}$$

The Engram module is integrated into the backbone via a residual connection: $\mathbf{H}^{(\ell)} \leftarrow \mathbf{H}^{(\ell)} + \mathbf{Y}$, followed by the standard Attention and MoE. Crucially, Engram is not applied to every layer; its specific placement is governed by the system-level latency constraints detailed in Section 2.5.

和 RULER (Hsieh et al.) 上显著优于基线（例如，Multi-Query NIAH：97.0 vs. 84.2；Variable Tracking：89.0 vs. 77.0）。

最后，我们将基础设施感知的效率确立为一等原则。与 MoE 的动态路由不同，Engram 采用确定性 ID 来实现运行时预取，使通信与计算重叠。实证结果表明，将一个 1000 亿参数的表卸载到主机内存所产生的开销可忽略不计（< 3%）。这证明 Engram 有效地绕过了 GPU 内存限制，促进了激进的参数扩展。

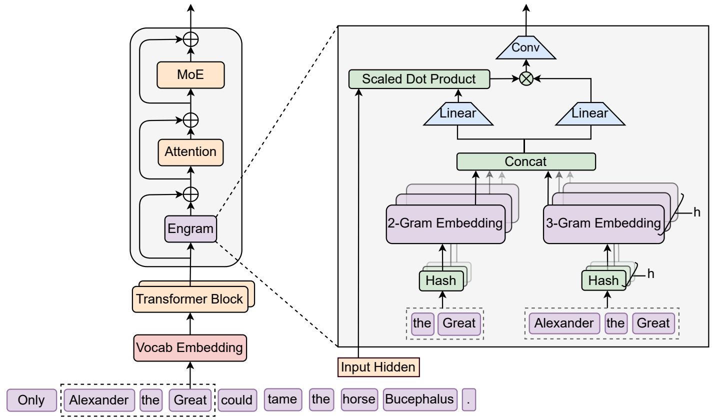
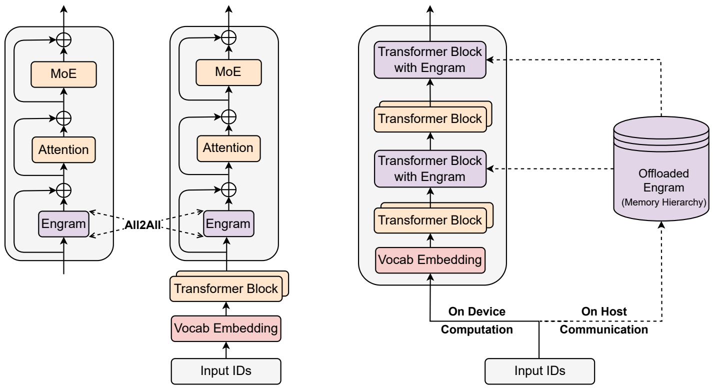


Figure 1 | **The Engram Architecture.** 该模块通过检索静态的 N-gram 记忆，并通过上下文感知门控将其与动态隐藏状态融合，从而增强主干网络。此模块仅应用于特定层，以将记忆与计算解耦，同时保持标准输入嵌入和解嵌入模块不变。

2. Architecture

2.1. Overview

如 Figure 1 所示，Engram 是一个条件记忆模块，旨在通过结构上将静态模式存储与动态计算分离来增强 Transformer 主干网络。形式上，给定输入序列 $X = (x_1, \ldots, x_T)$ 和层 ℓ 处的隐藏状态 $\mathbf{H}^{(\ell)} \in \mathbb{R}^{T \times d}$，该模块分两个功能阶段处理每个位置 t：**retrieval and fusion**。首先，如 节 2.2 中详述，我们提取并压缩后缀 N-gram，通过哈希确定性检索静态嵌入向量。随后，在 节 2.3 中，这些检索到的嵌入由当前隐藏状态动态调制，并通过轻量级卷积进行精炼。最后，我们在 节 2.4 中讨论与多分支架构的集成，并在 节 2.5 中讨论系统级设计。



(a) Engram at training **(b) Engram at inference**

Figure 2 | **System implementation of Engram.** (a) Training Phase: The massive embedding tables are sharded across available GPUs. An All-to-All communication primitive is employed to retrieve active embedding rows across devices. (b) Inference Phase: Engram tables are offloaded to host memory. By exploiting the deterministic retrieval logic, the host asynchronously prefetches and transfers embeddings, overlapping communication with the on-device computation of preceding Transformer blocks.

2.4. Integration with Multi-branch Architecture

In this work, rather than standard single-stream connections (He et al., 2016), we adopt the advanced multi-branch architecture as our default backbone, chosen for its superior modeling capabilities (Larsson et al., 2017; Szegedy et al., 2015; Xie et al., 2025; Zhu et al., 2025). A defining characteristic of this architecture is the expansion of the residual stream into M parallel branches, where information flow is modulated by learnable connection weights.

Although the Engram module is inherently topology-agnostic, adapting it to this multi-branch framework necessitates structural optimization to balance efficiency and expressivity. Specifically, we implement a parameter-sharing strategy: a single sparse embedding table and a Value projection matrix $\mathbf{W}_V$ are shared across all M branches, whereas M distinct Key projection matrices $\{\mathbf{W}_K^{(m)}\}_{m=1}^M$ are employed to enable branch-specific gating behaviors. For the m-th branch with hidden state $\mathbf{h}_t^{(m)}$, the branch-specific gating signal is computed as:

$$\alpha_t^{(m)} = \sigma\left(\frac{\mathrm{RMSNorm}(\mathbf{h}_t^{(m)})^\top \mathrm{RMSNorm}(\mathbf{W}_K^{(m)}\mathbf{e}_t)}{\sqrt{d}}\right). \tag{6}$$

The retrieved memory is then modulated by these independent gates applied to the shared value vector: $\mathbf{u}_t^{(m)} = \alpha_t^{(m)} \cdot (\mathbf{W}_V \mathbf{e}_t)$. This design allows the linear projections (one $\mathbf{W}_V$ and M distinct $\mathbf{W}_K^{(m)}$) to be fused into a single dense FP8 matrix multiplication, maximizing the compute utilization of modern GPUs. Unless otherwise stated, all experiments utilize this integration with Manifold-Constrained Hyper-Connections ($M = 4$) (Xie et al., 2025).

2.2. Sparse Retrieval via Hashed N-grams

第一阶段将局部上下文映射到静态记忆条目，涉及分词器压缩和通过确定性哈希检索嵌入。

Tokenizer Compression 虽然 N-gram 模型通常直接对分词器输出进行操作，但标准子词分词器优先考虑 *lossless reconstruction*，通常为语义上等价的术语分配不相交的 ID（例如，`Apple` 与 `␣apple`）(Kudo and Richardson, 2018; Li et al., 2023b)。为了最大化语义密度，我们实现了一个词汇表投影层。具体来说，我们预先计算一个满射函数 $\mathcal{P}: V \to V'$，该函数根据规范化的文本等价性（使用 NFKC (Whistler, 2025)、小写转换等）将原始词元 ID 折叠为规范标识符。在实践中，此过程对于 128k 分词器，有效词汇量大小减少了 23%（参见 Appendix C）。形式上，对于位置 t 的词元，我们将其原始 ID x_t 映射到规范 ID $x'_t = \mathcal{P}(x_t)$，以形成后缀 N-gram $g_{t,n} = (x'_{t-n+1}, \ldots, x'_t)$。

Multi-Head Hashing. 直接参数化所有可能的 N-gram 的组合空间是难以处理的。遵循 Tito Svenstrup et al. (2017)，我们采用基于哈希的方法。为了减轻冲突，我们为每个 N-gram 阶数 n 使用 K 个不同的哈希头。每个头 k 通过一个确定性函数 $\varphi_{n,k}$ 将压缩后的上下文映射到嵌入表 $\mathbf{E}_{n,k}$（大小为质数 $M_{n,k}$）内的一个索引：

$$z_{t,n,k} \triangleq \varphi_{n,k}(g_{t,n}), \quad \mathbf{e}_{t,n,k} = \mathbf{E}_{n,k}[z_{t,n,k}]. \tag{1}$$

在实践中，$\varphi_{n,k}$ 被实现为一个轻量级的乘法-XOR 哈希。我们通过连接所有检索到的嵌入来构建最终的记忆向量 $\mathbf{e}_t \in \mathbb{R}^{d_{\text{mem}}}$：

$$\mathbf{e}_t \triangleq \mathop{\Big\Vert}_{n=2}^{N} \mathop{\Big\Vert}_{k=1}^{K} \mathbf{e}_{t,n,k}. \tag{2}$$

2.3. Context-aware Gating

检索到的嵌入 $\mathbf{e}_t$ 作为上下文无关的先验。然而，由于是静态的，它们本质上缺乏上下文适应性，并且可能因哈希冲突或多义性 (Haber and Poesio, 2024) 而受到噪声影响。为了增强表达能力并解决这种歧义，我们采用了受注意力机制 (Bahdanau et al., 2015; Vaswani et al., 2017) 启发的上下文感知门控机制。具体来说，我们利用当前隐藏状态 $\mathbf{h}_t$——它已通过先前的注意力层聚合了全局上下文——作为动态查询，而检索到的记忆 $\mathbf{e}_t$ 则作为键和值投影的来源：

$$\mathbf{k}_t = \mathbf{W}_K \mathbf{e}_t, \quad \mathbf{v}_t = \mathbf{W}_V \mathbf{e}_t \tag{3}$$

其中 $\mathbf{W}_K, \mathbf{W}_V$ 是可学习的投影矩阵。为了确保梯度稳定性 (Dehghani et al., 2023)，我们在计算标量门 $\alpha_t \in (0, 1)$ 之前对查询和键应用 RMSNorm (Zhang and Sennrich, 2019)：

$$\alpha_t = \sigma\left(\frac{\mathrm{RMSNorm}(\mathbf{h}_t)^\top \mathrm{RMSNorm}(\mathbf{k}_t)}{\sqrt{d}}\right). \tag{4}$$

门控输出定义为 $\tilde{\mathbf{v}}_t = \alpha_t \cdot \mathbf{v}_t$。这种设计强制了语义对齐：如果检索到的记忆 $\mathbf{e}_t$ 与当前上下文 $\mathbf{h}_t$ 相矛盾，门 α_t 倾向于趋近于零，从而有效地抑制噪声。

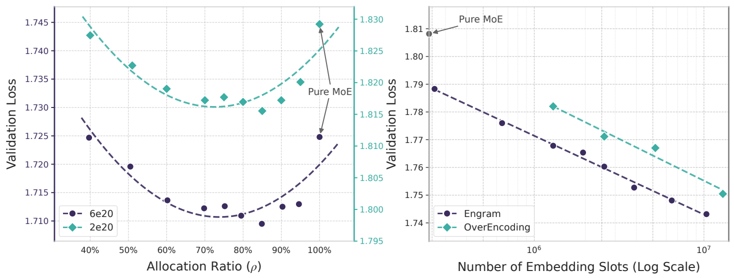


Figure 3 | **Sparsity allocation and Engram scaling.** Left: Validation loss across allocation ratios ρ. Two compute budgets are shown (2e20 and 6e20 FLOPs). Both regimes exhibit a U-shape, with hybrid allocation surpassing Pure MoE. Right: Scaling behavior in the infinite-memory regime. Validation loss exhibits a log-linear trend with respect to the number of embeddings.

2.5. System Efficiency: Decoupling Compute and Memory

Scaling memory-augmented models is often constrained by the limited capacity of GPU high-bandwidth memory (HBM). However, the deterministic retrieval mechanism of Engram naturally supports the decoupling of parameter storage from computational resources. Unlike MoE, which relies on runtime hidden states for dynamic routing, Engram's retrieval indices depend solely on the input token sequence. This predictability facilitates specialized optimization strategies for both training and inference, as illustrated in Figure 2.

During training, to accommodate large-scale embedding tables, we employ standard model parallelism by sharding the tables across available GPUs. An All-to-All communication primitive is used to gather active rows in the forward pass and dispatch gradients in the backward pass, enabling the total memory capacity to scale linearly with the number of accelerators.

During inference, this deterministic nature enables a prefetch-and-overlap strategy. Since memory indices are known prior to the forward pass, the system can asynchronously retrieve embeddings from abundant host memory via PCIe. To effectively mask communication latency, the Engram module is placed at specific layers within the backbone, leveraging the computation of preceding layers as a buffer to prevent GPU stalls. This necessitates a hardware-algorithm co-design strategy: while placing Engram deeper extends the compute window available for hiding latency, our ablation in Section 6.2 shows that modeling performance favors early intervention to offload local pattern reconstruction. Therefore, the optimal placement must simultaneously satisfy both modeling and system latency constraints.

Furthermore, natural language N-grams inherently follow a Zipfian distribution (Chao and Zipf, 1950; Piantadosi, 2014), where a small fraction of patterns accounts for the vast majority of memory accesses. This statistical property motivates a Multi-Level Cache Hierarchy: frequently accessed embeddings can be cached in faster storage tiers (e.g., GPU HBM or Host DRAM), while the long tail of rare patterns resides in slower, high-capacity media (e.g., NVMe SSD). This stratification allows Engram to scale to massive memory capacities with minimal impact on effective latency.

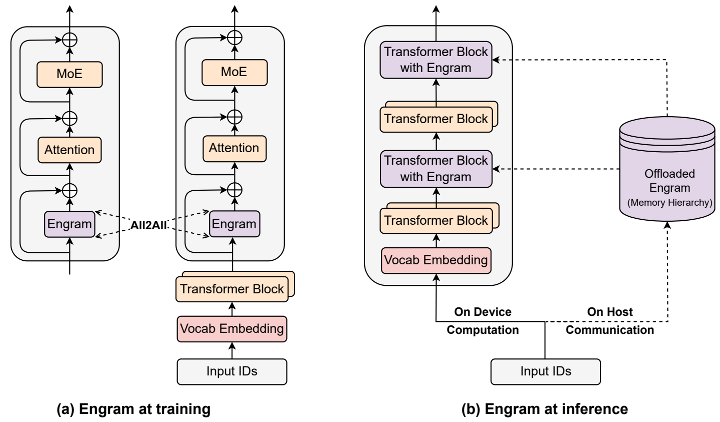


Figure 2 | **System implementation of Engram.** (a) 训练阶段：海量嵌入表被分片到可用的GPU 上。采用 All-to-All 通信原语来跨设备检索活跃的嵌入行。(b) 推理阶段：Engram 表被卸载到主机内存。通过利用确定性检索逻辑，主机异步预取并传输嵌入，使通信与设备上前一个Transformer 块的计算重叠。

最后，为了扩展感受野并增强模型的非线性，我们引入了一个短深度因果卷积 (Gu et al., 2022; Peng et al., 2023)。令 $\tilde{\mathbf{V}} \in \mathbb{R}^{T\times d}$ 表示门控值序列。使用核大小 w（设为 4）、膨胀率 δ（设为最大 N-gram 阶数）和 SiLU 激活函数 (Elfwing et al., 2018)，最终输出 $\mathbf{Y}$ 计算如下：

$$\mathbf{Y} = \text{SiLU}\left(\text{Conv1D}(\text{RMSNorm}(\tilde{\mathbf{V}}))\right) + \tilde{\mathbf{V}}, \tag{5}$$

Engram 模块通过残差连接集成到主干网络中：$\mathbf{H}^{(\ell)} \leftarrow \mathbf{H}^{(\ell)} + \mathbf{Y}$，随后是标准的注意力层和 MoE。至关重要的是，Engram 并不应用于每一层；其具体放置由 节 2.5 中详述的系统级延迟约束所控制。

2.4. Integration with Multi-branch Architecture

在本工作中，我们并未采用标准的单流连接 (He et al., 2016)，而是选择了先进的多分支架构作为默认主干网络，因其卓越的建模能力 (Larsson et al., 2017; Szegedy et al., 2015; Xie et al., 2025; Zhu et al., 2025)。该架构的一个显著特征是将残差流扩展为 M 个并行分支，其中信息流由可学习的连接权重进行调制。

尽管 Engram 模块本质上是拓扑无关的，但将其适配到多分支框架中需要进行结构优化，以平衡效率与表达能力。具体而言，我们实施了一种参数共享策略：一个稀疏嵌入表和一个值投影矩阵 $\mathbf{W}_V$ 在所有 M 个分支间共享，而 M 个不同的键投影矩阵 $\{\mathbf{W}_K^{(m)}\}_{m=1}^{M}$ 被用来实现分支特定

3. Scaling Laws and Sparsity Allocation

Engram, as an instantiation of conditional memory, is structurally complementary to the conditional computation provided by MoE experts. This section investigates the scaling properties of this duality and how to optimally allocate sparse capacity. Specifically, two key questions drive our research:

1. **Allocation under Finite Constraints.** When total parameters and training compute are fixed (Iso-parameter and Iso-FLOPs), how should we split the sparse capacity between MoE experts and Engram embeddings?
2. **Infinite Memory Regime.** Considering the non-scaling $O(1)$ overhead of Engram, if the memory budget is relaxed or scaled aggressively, what scaling behavior does Engram exhibit by itself?

3.1. Optimal Allocation Ratio Between MoE and Engram

Compute-matched formulation. We analyze the trade-off using three parameter metrics:

- P_{tot}: total trainable parameters, excluding vocabulary embedding and LM head.
- P_{act}: activated parameters per token. This quantity determines the training cost (FLOPs).
- $P_{\text{sparse}} \triangleq P_{\text{tot}} - P_{\text{act}}$: the *inactive* parameters, which represents the "free" parameter budget available for scaling model size without incurring computational cost (e.g., unselected experts or unretrieved embeddings).

We keep P_{tot} and P_{act} fixed within each FLOPs budget, so that models have the same number of parameters and the same per-token FLOPs. For MoE, P_{act} is determined by the top-k selected experts, while the parameters of non-selected experts contribute to P_{sparse}. For Engram, only a constant number of slots are retrieved per token, so scaling the number of embedding slots increases P_{tot} without increasing per-token FLOPs.

Allocation ratio. We define the allocation ratio $\rho \in [0, 1]$ as the fraction of the inactive-parameter budget assigned to MoE expert capacity:

$$P_{\text{MoE}}^{(\text{sparse})} = \rho\, P_{\text{sparse}}, \qquad P_{\text{Engram}} = (1-\rho)\, P_{\text{sparse}}. \tag{7}$$

Intuitively:

- $\rho = 1$ corresponds to a pure MoE model (all inactive parameters are routed experts).
- $\rho < 1$ reduces the number of routed experts and reallocates the freed parameters to Engram embedding slots.

Experimental protocol. We evaluate this trade-off at two compute regimes and maintain a constant sparsity ratio $P_{\text{tot}}/P_{\text{act}} \approx 10$ across both settings:

- $C = 2 \times 10^{20}$ FLOPs: $P_{\text{tot}} \approx 5.7\text{B}$ and $P_{\text{act}} = 568\text{M}$. The baseline ($\rho = 1$) has a total of 106 experts.
- $C = 6 \times 10^{20}$ FLOPs: $P_{\text{tot}} \approx 9.9\text{B}$ and $P_{\text{act}} = 993\text{M}$. The baseline ($\rho = 1$) has a total of 99 experts.

For different ρ, we construct the corresponding model only by adjusting the number of routed experts and the number of Engram embedding slots. All runs use the identical training pipeline and optimization hyperparameters.

的门控行为。对于第 m 个具有隐藏状态 $\mathbf{h}_t^{(m)}$ 的分支，其分支特定的门控信号计算如下：

$$\alpha_t^{(m)} = \sigma\left(\frac{\text{RMSNorm}(\mathbf{h}_t^{(m)})^\top \text{RMSNorm}(\mathbf{W}_K^{(m)}\mathbf{e}_t)}{\sqrt{d}}\right). \tag{6}$$

检索到的记忆随后由这些独立门控信号调制共享的值向量：$\mathbf{u}_t^{(m)} = \alpha_t^{(m)} \cdot (\mathbf{W}_V \mathbf{e}_t)$。该设计允许线性投影（一个 $\mathbf{W}_V$ 和 M 个不同的 $\mathbf{W}_K^{(m)}$）融合为单个密集的 FP8 矩阵乘法，从而最大化现代 GPU 的计算利用率。除非另有说明，所有实验均采用这种与流形约束超连接 ($M = 4$) (Xie et al., 2025) 的集成方式。

2.5. System Efficiency: Decoupling Compute and Memory

扩展记忆增强模型通常受到 GPU 高带宽内存（HBM）有限容量的制约。然而，Engram 的确定性检索机制天然支持将参数存储与计算资源解耦。与依赖运行时隐藏状态进行动态路由的 MoE 不同，Engram 的检索索引仅依赖于输入词元序列。这种可预测性为训练和推理都提供了专门的优化策略，如Figure 2所示。

在训练期间，为了容纳大规模的嵌入表，我们采用标准的模型并行方法，将表分片到可用的 GPU 上。使用 All-to-All 通信原语在前向传播中收集活跃行，并在反向传播中分发梯度，从而使总内存容量能够随加速器数量线性扩展。

在推理期间，这种确定性特性使得预取与重叠策略成为可能。由于内存索引在前向传播之前已知，系统可以通过 PCIe 异步地从充裕的主机内存中检索嵌入。为了有效掩盖通信延迟，Engram 模块被放置在骨干网络中的特定层，利用前序层的计算作为缓冲区来防止 GPU 停顿。这需要一种硬件-算法协同设计策略：虽然将 Engram 放置得更深可以延长用于隐藏延迟的计算窗口，但我们在节 5.2中的消融实验表明，建模性能倾向于早期介入以卸载局部模式重建。因此，最优的放置必须同时满足建模和系统延迟的约束。

此外，自然语言的 N-元组天然遵循齐夫分布 (Chao and Zipf, 1950; Piantadosi, 2014)，其中一小部分模式占据了绝大多数内存访问。这一统计特性催生了多级缓存层次结构：频繁访问的嵌入可以缓存在更快的存储层级（例如 GPU HBM 或主机 DRAM）中，而长尾的罕见模式则驻留在速度较慢、容量较大的介质（例如 NVMe SSD）中。这种分层使得 Engram 能够扩展到海量内存容量，同时对有效延迟的影响最小。

3. Large Scale Pre-training

基于所提出的 Engram 架构和经验推导的分配定律，我们将 Engram 扩展到数十亿参数规模，以验证其在现实世界语言模型预训练中的有效性。具体而言，我们训练了四个模型：(1) **Dense-4B** (总计 41 亿参数), (2) **MoE-27B** (总计 267 亿参数), (3) **Engram-27B** (总计 267 亿参数), 以及 (4) **Engram-40B** (总计 395 亿参数)。所有模型均使用相同的数据课程（相同的 token 预算和顺序）进行训练，并且在激活参数数量上严格匹配。

Results and Analysis. Figure 3 (left) reveals a consistent U-shaped relationship between validation loss and the allocation ratio ρ. Remarkably, the Engram model achieves comparable performance to the pure MoE baseline ($\rho = 100\%$) even when the MoE allocation is reduced to just $\rho \approx 40\%$ (i.e., a total of 46 experts for the 5.7B model and 43 experts for the 9.9B model). Furthermore, the pure MoE baseline proves suboptimal: reallocating roughly 20%–25% of the sparse parameter budget to Engram yields the best performance. Quantitatively, in the 10B regime ($C = 6 \times 10^{20}$), validation loss improves from 1.7248 (at $\rho = 100\%$) to 1.7109 near the optimum of $\rho \approx 80\%$ ($\Delta = 0.0139$). Crucially, the location of this optimum is stable across regimes ($\rho \approx 75\%$–80%), suggesting a robust allocation preference across the examined scales (under fixed sparsity). This observed U-shape confirms the structural complementarity between the two modules:

- **MoE-dominated ($\rho \to 100\%$):** The model lacks dedicated memory for static patterns, forcing it to inefficiently reconstruct them through depth and computation.
- **Engram-dominated ($\rho \to 0\%$):** The model loses conditional computation capacity, hurting tasks that require dynamic, context-dependent reasoning; memory cannot replace computation in this regime.

3.2. Engram under Infinite Memory Regime

In Section 3.1, we optimized the allocation under a fixed parameter budget. We now explore the complementary setting: aggressive memory scaling. This investigation is motivated by Engram's unique ability to decouple storage from compute detailed in Section 2.5.

Experimental protocol. We utilize a fixed MoE backbone with $P_{\text{tot}} \approx$ 3B and $P_{\text{act}} =$ 568M, trained for 100B tokens to ensure convergence. On top of this backbone, we attach an Engram table and sweep the number of slots M from 2.58×10^5 to 1.0×10^7 (adding up to $\approx$ 13 billion parameters). For baselines, we compare against OverEncoding (Huang et al., 2025a), which integrates N-gram embeddings via averaging with the vocabulary embedding. We note that while other work such as SCONE (Yu et al., 2025) also investigates large-scale embeddings, it is primarily inference-focused and includes extra module (*f-gram model*) and additional training FLOPs, rendering it incompatible with the strict iso-compute constraints of this study.

Results. Figure 3 (right) demonstrates that scaling the number of memory slots yields a clear and consistent improvement in validation loss. Across the explored range, the curve follows a strict power law (linear in log-space), indicating that Engram provides a predictable scaling knob: larger memory continues to pay off without requiring additional computation. Crucially, regarding scaling efficiency: while the direct averaging approach of OverEncoding benefits from larger memory tables, Engram unlocks much larger scaling potential from the same memory budget. Together with the allocation law in Section 3.1, these results validate that conditional memory serves as a distinct, scalable axis of sparse capacity that complements the conditional computation of MoE.

4. Large Scale Pre-training

With the proposed Engram architecture and the empirically derived allocation law, we scale Engram to the multi-billion parameter to validate its efficacy in real-world language model pre-training. Specifically, we train four models: (1) **Dense-4B** (4.1B total parameters), (2) **MoE-27B**

Table 1 | **Pre-training performance comparison between dense, MoE, and Engram models**。所有模型均训练了 262B 个词元，并在激活参数量（3.8B）上匹配。Engram-27B 通过将参数从路由专家（72 → 55）重新分配到 5.7B 参数的 Engram 记忆，实现了与 MoE-27B 的等参数量。Engram-40B 在保持激活参数量预算固定的同时，进一步增加了 Engram 记忆（18.5B 参数）。完整的训练时间基准轨迹见Appendix B。

	Benchmark (Metric)	# Shots	Dense-4B	MoE-27B	Engram-27B	Engram-40B
	# Total Params		4.1B	26.7B	26.7B	39.5B
	# Activated (w/o token embed)		3.8B	3.8B	3.8B	3.8B
	# Trained Tokens		262B	262B	262B	262B
	# Experts (shared + routed, top-k)		-	2+72 (top-6)	2+55 (top-6)	2+55 (top-6)
	# Engram Params		-	-	5.7B	18.5B
Language Modeling	Pile (loss)	-	2.091	1.960	**1.950**	1.942
	Validation Set (loss)	-	1.768	1.634	**1.622**	1.610
Knowledge & Reasoning	MMLU (Acc.)	5-shot	48.6	57.4	**60.4**	60.6
	MMLU-Redux (Acc.)	5-shot	50.7	60.6	**64.0**	64.5
	MMLU-Pro (Acc.)	5-shot	21.1	28.3	**30.1**	31.3
	CMMLU (Acc.)	5-shot	47.9	57.9	**61.9**	63.4
	C-Eval (Acc.)	5-shot	46.9	58.0	**62.7**	63.3
	AGIEval (Acc.)	0-shot	29.1	38.6	**41.8**	45.9
	ARC-Easy (Acc.)	25-shot	76.8	86.5	**89.0**	90.1
	ARC-Challenge (Acc.)	25-shot	59.3	70.1	**73.8**	76.4
	TriviaQA (EM)	5-shot	33.0	48.8	**50.7**	51.8
	TriviaQA-ZH (EM)	5-shot	62.8	74.8	**76.3**	77.9
	PopQA (EM)	15-shot	15.1	19.2	**19.4**	21.2
	CCPM (Acc.)	0-shot	72.2	79.6	**87.1**	87.7
	BBH (EM)	3-shot	42.8	50.9	**55.9**	57.5
	HellaSwag (Acc.)	0-shot	64.3	71.8	**72.7**	73.1
	PIQA (Acc.)	0-shot	63.8	71.9	**73.5**	76.5
	WinoGrande (Acc.)	5-shot	64.0	67.6	**67.8**	68.1
Reading Comprehension	DROP (F1)	1-shot	41.6	55.7	**59.0**	60.7
	RACE-Middle (Acc.)	5-shot	72.4	80.9	**82.8**	83.3
	RACE-High (Acc.)	5-shot	66.0	75.4	**78.2**	79.2
	C3 (Acc.)	0-shot	57.7	60.1	**63.6**	61.8
Code & Math	HumanEval (Pass@1)	0-shot	26.8	37.8	**40.8**	38.4
	MBPP (Pass@1)	3-shot	35.4	46.6	**48.2**	46.2
	CruxEval-i (EM)	0-shot	27.6	30.7	**32.2**	36.2
	CruxEval-o (EM)	0-shot	28.7	34.1	**35.0**	35.3
	GSM8K (EM)	8-shot	35.5	58.4	**60.6**	62.6
	MGSM (EM)	8-shot	27.0	46.8	**49.4**	52.4
	MATH (EM)	4-shot	15.2	28.3	**30.7**	30.6

Table 1 | **Pre-training performance comparison between dense, MoE, and Engram models**. All models are trained for 262B tokens and are matched in activated parameters (3.8B). Engram-27B is iso-parameters with MoE-27B by reallocating parameters from routed experts (72 → 55) to a 5.7B-parameter Engram memory. Engram-40B further increases Engram memory (18.5B parameters) while keeping the activated-parameter budget fixed. Full training-time benchmark trajectories are reported in Appendix B.

	Benchmark (Metric)	# Shots	Dense-4B	MoE-27B	Engram-27B	Engram-40B
	# Total Params		4.1B	26.7B	26.7B	39.5B
	# Activated (w/o token embed)		3.8B	3.8B	3.8B	3.8B
	# Trained Tokens		262B	262B	262B	262B
	# Experts (shared + routed, top-k)		-	2 + 72 (top-6)	2 + 55 (top-6)	2 + 55 (top-6)
	# Engram Params		-	-	5.7B	18.5B
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	C-Eval (Acc.)	5-shot	46.9	58.0	**62.7**	63.3
	AGIEval (Acc.)	0-shot	29.1	38.6	**41.8**	45.9
	ARC-Easy (Acc.)	25-shot	76.8	86.5	**89.0**	90.1
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	PopQA (EM)	15-shot	15.1	19.2	**19.4**	21.2
	CCPM (Acc.)	0-shot	72.2	79.6	**87.1**	87.7
	BBH (EM)	3-shot	42.8	50.9	**55.9**	57.5
	HellaSwag (Acc.)	0-shot	64.3	71.8	**72.7**	73.1
	PIQA (Acc.)	0-shot	63.8	71.9	**73.5**	76.5
	WinoGrande (Acc.)	5-shot	64.0	67.6	**67.8**	68.1
Reading Comprehension	DROP (F1)	1-shot	41.6	55.7	**59.0**	60.7
	RACE-Middle (Acc.)	5-shot	72.4	80.9	**82.8**	83.3
	RACE-High (Acc.)	5-shot	66.0	75.4	**78.2**	79.2
	C3 (Acc.)	0-shot	57.7	60.1	**63.6**	61.8
Code & Math	HumanEval (Pass@1)	0-shot	26.8	37.8	**40.8**	38.4
	MBPP (Pass@1)	3-shot	35.4	46.6	**48.2**	46.2
	CruxEval-i (EM)	0-shot	27.6	30.7	**32.2**	36.2
	CruxEval-o (EM)	0-shot	28.7	34.1	**35.0**	35.3
	GSM8K (EM)	8-shot	35.5	58.4	**60.6**	62.6
	MGSM (EM)	8-shot	27.0	46.8	**49.4**	52.4
	MATH (EM)	4-shot	15.2	28.3	**30.7**	30.6

(26.7B total parameters), (3) **Engram-27B** (26.7B total parameters), and (4) **Engram-40B** (39.5B total parameters). All models are trained using an identical data curriculum (same token budget and order) and are strictly matched in the number of activated parameters.

4.1. Experimental Setup

Training Data and Model Configurations All models are pre-trained on a corpus of 262 billion tokens and we utilize the tokenizer from DeepSeek-v3 (Liu et al., 2024a) with a vocabulary size of 128k. For modeling, to ensure a controlled comparison, we adhere to a consistent default setting across all models unless explicitly stated otherwise. We utilize a 30-block Transformer with a

3.1. Experimental Setup

Training Data and Model Configurations 所有模型均在包含 2620 亿个标记的语料库上进行预训练，我们使用来自 DeepSeek-v3 (Liu et al., 2024a) 的分词器，其词汇表大小为 128k。在建模方面，为确保受控比较，除非另有明确说明，我们在所有模型中均遵循一致的默认设置。我们使用一个包含 30 个块的 Transformer，其隐藏层大小为 2560。每个块集成了一个具有 32 个头的多头潜在注意力（MLA）（DeepSeek-AI et al., 2024），并通过扩展率为 4 的 mHC (Xie et al., 2025) 连接到 FFN。所有模型均使用 Muon (Jordan et al., 2024; Team et al., 2025) 进行优化；详细的超参数列于Appendix A中。我们实例化了四个不同的模型：

- **Dense-4B** 作为基线模型。它采用上述骨干架构，在每个块中集成了一个标准密集 FFN。
- **MoE-27B** 将标准密集 FFN 替换为 DeepSeekMoE 模块 (Dai et al., 2024)。该模型配置了 72 个路由专家和 2 个共享专家（每个标记激活前 $k = 6$ 个路由专家），总参数量扩展至 26.7B，同时保持与 Dense-4B 相同的激活参数数量。
- **Engram-27B** 严格源自 **MoE-27B** 架构以确保公平比较。我们将路由专家数量从 72 个减少到 55 个，并将释放出的参数重新分配到一个 5.7B 参数的嵌入模块 ($\rho = 74.3\%$)，使模型总大小保持在 26.7B 不变。关于 Engram 配置，我们在第 2 层和第 15 层实例化该模块，并将最大 N-gram 大小设置为 3，头数设置为 8，维度设置为 1280。在优化方面，嵌入参数使用 Adam (Kingma, 2014) 进行更新，学习率按 5× 缩放且无权重衰减，而卷积参数初始化为零，以在训练开始时严格保持恒等映射。
- **Engram-40B** 保持与 Engram-27B 相同的骨干架构和计算预算，但将稀疏嵌入模块扩展到 18.5B 参数（总计 39.5B 参数）。该模型旨在研究 Engram 的缩放特性。

Evaluation Protocol 我们在涵盖语言建模、知识、推理、阅读理解以及代码/数学的多样化基准测试套件上评估模型。对于每个基准测试，我们遵循标准的提示协议和评估指标。

- **Language Modeling:** 我们报告 The Pile (Gao et al., 2020) 测试集上的损失，以及一个从与训练数据相同分布中抽取的验证集上的损失。
- **Knowledge & Reasoning:** MMLU (Hendrycks et al., 2021a), MMLU-Redux (Gema et al., 2025), MMLU-Pro (Wang et al., 2024b), CMMLU (Li et al., 2024), C-Eval (Huang et al., 2023), AGIEval (Zhong et al., 2024), ARC-Easy/Challenge (Clark et al., 2018), TriviaQA (Joshi et al., 2017), TriviaQA-ZH（内部）, PopQA (Mallen et al., 2023), CCPM (Li et al., 2021), BBH (Suzgun et al., 2023), HellaSwag (Zellers et al., 2019), PIQA (Bisk et al., 2020), 以及 WinoGrande (Sakaguchi et al., 2021)。
- **Reading Comprehension:** DROP (Dua et al., 2019), RACE (Middle/High) (Lai et al., 2017), 以及 C3 (Sun et al., 2020)。
- **Code & Math:** HumanEval (Chen et al., 2021), MBPP (Austin et al., 2021), CruxEval (Gu et al., 2024), GSM8K (Cobbe et al., 2021), MGSM (Shi et al., 2023), 以及 MATH (Hendrycks et al., 2021b)。

hidden size of 2560. Each block integrates a Multi-head Latent Attention (MLA) (DeepSeek-AI et al., 2024) with 32 heads, connected to FFNs via mHC (Xie et al., 2025) with an expansion rate of 4. All models are optimized using Muon (Jordan et al., 2024; Team et al., 2025); detailed hyperparameters are listed in the Appendix A. We instantiate four distinct models:

- **Dense-4B** serves as the baseline model. It utilizes the backbone architecture described above, incorporating a standard dense FFN into every block.
- **MoE-27B** replaces the standard dense FFN with a DeepSeekMoE module (Dai et al., 2024). Configured with 72 routed experts and 2 shared experts (activating the top-$k = 6$ routed experts per token), this model scales to 26.7B total parameters while maintaining the same activated parameters as Dense-4B.
- **Engram-27B** is strictly derived from the **MoE-27B** architecture to ensure fair comparison. We reduce the number of routed experts from 72 to 55 and reallocate the freed parameters to a 5.7B-parameter embedding module ($\rho = 74.3\%$), keeping the total model size constant at 26.7B. Regarding the Engram configuration, we instantiate the module at layers 2 and 15 and set the maximum N-gram size to 3, the number of heads to 8, and the dimension to 1280. For optimization, the embedding parameters are updated using Adam (Kingma, 2014) with a learning rate scaled by $5\times$ and no weight decay, while the convolution parameters are initialized to zero to strictly preserve the identity mapping at the start of training.
- **Engram-40B** retains the same backbone and computation budget as Engram-27B but scales the sparse embedding module to 18.5B parameters (totaling 39.5B parameters). This model is designed to investigate the scaling properties of Engram.

Evaluation Protocol We evaluate models on a diverse suite of benchmarks spanning language modeling, knowledge, reasoning, reading comprehension, and code/math. For each benchmark, we follow standard prompting protocols and evaluation metrics.

- **Language Modeling:** We report loss on the test set of The Pile (Gao et al., 2020) and an validation set drawn from the same distribution as the training data.
- **Knowledge & Reasoning:** MMLU (Hendrycks et al., 2021a), MMLU-Redux (Gema et al., 2025), MMLU-Pro (Wang et al., 2024b), CMMLU (Li et al., 2024), C-Eval (Huang et al., 2023), AGIEval (Zhong et al., 2024), ARC-Easy/Challenge (Clark et al., 2018), TriviaQA (Joshi et al., 2017), TriviaQA-ZH (internal), PopQA (Mallen et al., 2023), CCPM (Li et al., 2021), BBH (Suzgun et al., 2023), HellaSwag (Zellers et al., 2019), PIQA (Bisk et al., 2020), and WinoGrande (Sakaguchi et al., 2021).
- **Reading Comprehension:** DROP (Dua et al., 2019), RACE (Middle/High) (Lai et al., 2017), and C3 (Sun et al., 2020).
- **Code & Math:** HumanEval (Chen et al., 2021), MBPP (Austin et al., 2021), CruxEval (Gu et al., 2024), GSM8K (Cobbe et al., 2021), MGSM (Shi et al., 2023), and MATH (Hendrycks et al., 2021b).

4.2. Experimental Results

Table 1 summarizes the main results. First, consistent with prior literature (Borgeaud et al., 2022; He, 2024; Shazeer et al., 2017), sparse architectures demonstrate superior scaling laws compared to dense models. Under the same training compute budget, all three sparse variants (MoE-27B, Engram-27B/40B) significantly outperform the iso-FLOPs Dense-4B baseline across all benchmarks.

Table 2 | **Long-context performance comparison.** 括号内的数值（例如 *(50k, 1.62)*）表示长上下文扩展前的预训练步数及相应的损失。两个关键发现：(1) 仅使用基线预训练 FLOPs 的 82%（41k 对比 50k），Engram-27B 在 LongPPL (Fang et al.) 性能上与基线持平，同时在 RULER (Hsieh et al.) 上取得了显著更高的准确率；(2) 在等预训练损失（46k）和等预训练 FLOPs（50k）两种设置下，Engram-27B 在所有指标上都大幅优于基线。**Bold** 表示最佳，underline 表示次佳。

Model	LongPPL (32k)				RULER (32k)							
	Perplexity (↓)				NIAH Accuracy (↑)				Other Tasks (↑)			
	Book	Paper	Code	L-CoT	S	MK	MV	MQ	VT	CWE	FWE	QA
MoE-27B *(50k, 1.63)*	4.38	2.91	2.49	14.16	**100.0**	88.0	92.7	84.2	77.0	<u>4.5</u>	73.0	34.5
=== Ours (Engram Variants) === Engram-27B *(46k, 1.63)*	<u>4.19</u>	<u>2.84</u>	<u>2.45</u>	<u>13.59</u>	97.6	<u>89.0</u>	<u>95.5</u>	**97.0**	<u>87.2</u>	4.3	<u>98.6</u>	37.5
Engram-27B *(50k, 1.62)*	**4.14**	**2.82**	**2.44**	**13.41**	99.3	**89.3**	**96.5**	**97.0**	**89.0**	**5.9**	**99.3**	<u>40.5</u>

3.2. Experimental Results

表 1 总结了主要结果。首先，与先前文献 (Borgeaud et al., 2022; He, 2024; Shazeer et al., 2017) 一致，稀疏架构展现出比密集模型更优的缩放定律。在相同的训练计算预算下，所有三种稀疏变体（MoE-27B、Engram-27B/40B）在所有基准测试上都显著优于等 FLOPs 的 Dense-4B 基线。

更重要的是，Engram-27B 持续优于等参数量和等 FLOPs 的 MoE-27B 基线。有趣的是，这些收益并不局限于知识密集型任务（例如，MMLU：+3.0，MMLU-Pro：+1.8，CMMLU：+4.0），尽管在这些任务中，记忆容量直观上是有益的。我们在通用推理领域（例如，BBH：+5.0，ARC-Challenge：+3.7，DROP：+3.3）以及代码和数学推理领域（例如，HumanEval：+3.0，MBPP：+1.6，GSM8K：+2.2，MATH：+2.4）观察到了更显著的改进。为了减少基准测试噪声的影响并可视化训练动态，我们在Appendix B中提供了预训练期间完整的基准测试轨迹。这些结果支持了我们的假设：引入一个专用的知识查找原语，其表征效率的提升超越了将整个稀疏预算全部分配给条件计算所能达到的效果。

最后，扩展到 Engram-40B 进一步降低了预训练损失，并在大多数基准测试上提升了性能。尽管它尚未在所有任务上严格超越 Engram-27B，但这很可能是训练不足导致的。我们观察到，在训练接近尾声时，Engram-40B 与基线之间的训练损失差距持续扩大，这表明在当前 token 预算内，扩展的记忆容量尚未完全饱和。

4. Long Context Training

通过将局部依赖建模卸载给静态查找，Engram 架构保留了宝贵的注意力容量以管理全局上下文。在本节中，我们通过进行长上下文扩展训练 (Gao et al., 2025; Peng et al., 2024)，从经验上验证了这一结构优势。通过一个严格的评估协议，该协议将架构贡献与基础模型能力分离开来，我们证明了 Engram 在长程检索和推理任务中能带来显著增益。

Table 2 | **Long-context performance comparison.** Parenthetical values (e.g. *(50k, 1.62)*) denote the pre-training steps and the corresponding loss prior to the long-context extension. Two key findings: (1) With only 82% of the pre-training FLOPs (41k vs. 50k), Engram-27B matches the baseline's LongPPL (Fang et al.) performance while achieving significantly higher accuracy on RULER (Hsieh et al.); (2) Under both iso-pretraining-loss (46k) and iso-pretraining-FLOPs (50k) settings, Engram-27B substantially outperforms the baseline across all metrics. **Bold** indicates the best and <u>underline</u> the second.

Model	**LongPPL (32k)**				**RULER (32k)**							
	Perplexity (↓)				NIAH Accuracy (↑)				Other Tasks (↑)			
	Book	Paper	Code	L-CoT	S	MK	MV	MQ	VT	CWE	FWE	QA
MoE-27B *(50k, 1.63)*	4.38	2.91	2.49	14.16	**100.0**	88.0	92.7	84.2	77.0	<u>4.5</u>	73.0	34.5
Engram-27B *(41k, 1.66)*	4.37	2.92	2.50	14.26	<u>99.6</u>	88.3	93.0	89.5	83.2	3.8	99.6	**44.0**
Engram-27B *(46k, 1.63)*	<u>4.19</u>	<u>2.84</u>	<u>2.45</u>	<u>13.59</u>	97.6	<u>89.0</u>	<u>95.5</u>	**97.0**	<u>87.2</u>	4.3	<u>98.6</u>	37.5
Engram-27B *(50k, 1.62)*	**4.14**	**2.82**	**2.44**	**13.41**	99.3	**89.3**	**96.5**	**97.0**	**89.0**	**5.9**	**99.3**	<u>40.5</u>

More importantly, Engram-27B consistently improves over the iso-parameter and iso-FLOPs MoE-27B baseline. Interestingly, these gains are not limited to knowledge-intensive tasks (e.g., MMLU: +3.0, MMLU-Pro: +1.8, CMMLU: +4.0), where memory capacity is intuitively beneficial. We observe even more significant improvements in general-reasoning domains (e.g., BBH: +5.0, ARC-Challenge: +3.7, DROP: +3.3), as well as code and mathematical reasoning (e.g., HumanEval: +3.0, MBPP: +1.6, GSM8K: +2.2, MATH: +2.4). To reduce the impact of benchmark noise and to visualize training dynamics, we provide full benchmark trajectories during pre-training in Appendix B. These results support our hypothesis that introducing a dedicated knowledge lookup primitive improves representation efficiency beyond what can be achieved by allocating the entire sparse budget to conditional computation.

Finally, scaling to Engram-40B further reduces pre-training loss and improves performance across most benchmarks. Although it does not yet strictly dominate Engram-27B on every task, this is likely an artifact of under-training. We observe that the training loss gap between Engram-40B and the baselines continues to widen towards the end of training, suggesting that the expanded memory capacity has not yet fully saturated within the current token budget.

5. Long Context Training

By offloading local dependency modeling to static lookups, the Engram architecture preserves valuable attention capacity for managing global context. In this section, we empirically verify this structural advantage by conducting long-context extension training (Gao et al., 2025; Peng et al., 2024). Through a rigorous evaluation protocol that isolates architectural contributions from base model capabilities, we demonstrate that Engram yields significant gains in long-range retrieval and reasoning tasks.

5.1. Experimental Setup

Training Details. To enable long-context capabilities, we adopt the context expansion strategy introduced in DeepSeek-V3 (Liu et al., 2024a). Following the pre-training stage, we apply YaRN (Peng et al., 2024) for context window extension in a 32768-token context training stage for 5,000 steps (30B tokens of high-quality, long-context data). The hyper-parameters are scale $s = 10, \alpha = 1, \beta = 32$ and the scaling factor $f = 0.707$.

4.1. Experimental Setup

Training Details. 为了启用长上下文能力，我们采用了 DeepSeek-V3 (Liu et al., 2024a) 中引入的上下文扩展策略。在预训练阶段之后，我们应用 YaRN (Peng et al., 2024) 进行上下文窗口扩展，在一个 32768 个标记的上下文训练阶段中进行 5000 步训练（使用 300 亿个标记的高质量长上下文数据）。超参数为缩放尺度 $s = 10, \alpha = 1, \beta = 32$ 和缩放因子 $f = 0.707$。

Model Configurations. 我们比较了四种不同模型配置下的上下文扩展效果。我们使用了 MoE-27B 和 Engram-27B 的最终预训练检查点（50k 步）。此外，为了严格评估架构效率，我们为 Engram-27B 选择了两个中间检查点，分别在 41k 步和 46k 步。尽管初始阶段不同，所有变体都遵循 *the exact same* 上下文扩展训练协议。关键的是，选择 Engram-27B（46k）是因为其预训练损失与完全训练的 MoE-27B（50k）相同。这创建了一个受控的" 等损失" 设置，确保在上下文扩展期间出现的任何性能差异都可归因于架构，而非模型的起始质量。

Evaluation Benchmarks. 我们使用 LongPPL (Fang et al.) 和 RULER (Hsieh et al.) 来评估长上下文性能。对于 LongPPL，我们构建了涵盖四个类别的评估集：长书籍、研究论文、代码仓库和长思维链轨迹。对于 RULER，我们在 14 个子集上进行评估，这些子集聚合为 8 个类别：单针（S）、多键（MK）、多值（MV）和多查询（MQ）大海捞针；多跳变量追踪（VT）、常见词提取（CWE）、高频词提取（FWE）和问答（QA）。

4.2. Experimental Results

评估结果总结于 表 3.2。为了准确评估印迹架构的贡献，我们的分析分两步进行：首先，将基础模型能力的影响与架构设计解耦；其次，进行受控分析。

1. Long-Context Capability Beyond Attention Mechanics. 虽然注意力机制和位置编码为上下文处理提供了结构基础 (Press et al., 2022; Su et al., 2024; Xiao et al., 2024; Yang et al., 2025)，但我们的结果表明，长上下文性能并非仅由架构先验决定。观察印迹模型的轨迹（41k → 50k），我们发现长上下文性能随着预训练的推进而单调提升，即使在控制模型架构相同且在上下文扩展阶段计算预算固定的情况下也是如此。这表明长上下文性能与基础模型的通用建模能力存在内在耦合。因此，严格的架构比较必须通过对齐基础模型损失而非仅仅对齐训练步数来控制这一混杂变量。

2. Architectural Superiority under Controlled Settings. 基于上述原则，我们将印迹模型与 MoE 基线进行基准测试。在控制基础能力的情况下，印迹模块的效率优势变得明显：

- **Iso-Loss Setting (46k vs. Baseline):** 此设置严格隔离了架构效率。当比较印迹-27B（46k）与完全训练的 MoE-27B（50k）——这些模型在预训练损失上对齐——时，印迹模型显示出显著优势。具体而言，它在复杂检索任务上优于基线（例如，多查询 NIAH：97.0 对比 84.2；VT：87.2 对比 77.0）。

Model Configurations. We compare context extensions across four distinct model configurations. We utilize the final pre-training checkpoints (50k steps) for both MoE-27B and Engram-27B. Additionally, to rigorously benchmark architectural efficiency, we select two intermediate checkpoints for Engram-27B at 41k and 46k steps. Despite differing initialization stages, all variants undergo *the exact same* context extension training protocol. Crucially, Engram-27B (46k) is selected because it exhibits the same pre-training loss as the fully trained MoE-27B (50k). This creates a controlled "Iso-Loss" setting, ensuring that any performance divergence during context extension is attributable to the architecture rather than the starting quality of the model.

Evaluation Benchmarks. We assess long-context performance using LongPPL (Fang et al.) and RULER (Hsieh et al.). For LongPPL, we construct evaluation sets spanning four categories: long books, research papers, code repositories, and long chain-of-thought (CoT) trajectories. For RULER, we evaluate on 14 subsets aggregated into 8 categories: Single (S), Multi-keys (MK), Multi-values (MV) and Multi-queries (MQ) Needle-in-a-Haystack; Multi-hop Variable Tracking (VT), Common Words Extraction (CWE), Frequent Words Extraction (FWE), and Question Answering (QA).

5.2. Experimental Results

The evaluation results are summarized in Table 2. To accurately assess the contribution of the Engram architecture, our analysis proceeds in two steps: first, decoupling the impact of base model capability from architectural design, and second, conducting a controlled analysis.

1. Long-Context Capability Beyond Attention Mechanics. While attention mechanisms and positional encoding provide the structural basis for context processing (Press et al., 2022; Su et al., 2024; Xiao et al., 2024; Yang et al., 2025), our results indicate that long-context performance is not solely determined by architectural priors. Observing the trajectory of Engram (41k → 50k), we find that long-context performance improves monotonically with pre-training progression, even when controlling for identical model architecture and a fixed computational budget during the context extension stage. This suggests that long-context performance is intrinsically coupled with the general modeling ability of the base model. Consequently, a rigorous architectural comparison must control for this confounding variable by aligning base model loss, rather than merely aligning training steps.

2. Architectural Superiority under Controlled Settings. Guided by the principle above, we benchmark Engram against the MoE baseline. When controlling for base capability, the efficiency gains of the Engram module become evident:

- **Iso-Loss Setting (46k vs. Baseline):** This setting strictly isolates architectural efficiency. When comparing Engram-27B (46k) against the fully trained MoE-27B (50k)—models aligned on pre-training loss—Engram demonstrates significant gains. Specifically, it outperforms the baseline on complex retrieval tasks (e.g., Multi-Query NIAH: 97.0 vs. 84.2; VT: 87.2 vs. 77.0).
- **Iso-FLOPs Setting (50k vs. Baseline):** Under the standard iso-compute budget, Engram-27B (50k) further widens this gap, establishing the highest performance across the board.
- **Extreme Setting (≈ 82% Compute):** Even the early-stopped Engram-27B (41k) remains highly competitive against the fully trained MoE-27B (50k). It matches the baseline on LongPPL and surpasses it on RULER, underscoring the intrinsic superiority of the Engram architecture.

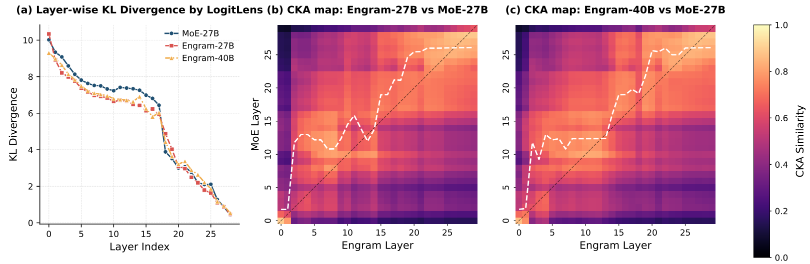

Figure 3 | **Analysis of representational alignment and convergence speed.** (a) 通过 LogitLens (nostalgebraist, 2020) 计算的逐层 KL 散度。早期层持续较低的散度表明 Engram 加速了预测收敛。(b-c) 由 CKA (Kornblith et al., 2019) 计算的相似性热力图。高相似性对角线的明显上移表明，Engram 的浅层在功能上等同于 MoE 模型的深层，从而有效地增加了模型的深度。

- **Iso-FLOPs Setting (50k vs. Baseline):** 在标准的等计算预算下，印迹-27B（50k）进一步扩大了这一差距，在所有方面都确立了最高性能。
- **Extreme Setting (≈ 82% Compute):** 即使是提前停止训练的印迹-27B（41k），与完全训练的 MoE-27B（50k）相比也仍然极具竞争力。它在 LongPPL 上与基线持平，并在 RULER 上超越基线，突显了印迹架构的内在优越性。

5. Analysis

在本节中，我们探究 Engram 的内部机制，包括其有效深度 (节 5.1)、核心模块设计 (节 5.2) 和参数敏感性 (节 5.3)。此外，我们评估了带卸载的推理吞吐量 (节 5.4)，并以一个案例研究 (节 5.5) 作为总结。

5.1. Is Engram functionally equivalent to increasing the model’s depth?

当前的 LLM 缺乏专用的知识查找原语，它们依赖计算来模拟记忆检索。如表 3所示，为了识别实体 "Diana, Princess of Wales"，LLM 必须消耗多层注意力和前馈网络来逐步组合特征 (Ghandeharioun et al., 2024; Jin et al., 2025; Li and Subramani, 2025)，这一过程理论上可以通过知识查找操作来识别。

基于此，我们提出，通过为模型配备显式的知识查找能力，Engram 通过免除模型早期的特征组合阶段，有效地模拟了模型深度的增加。为了验证这一假设，我们使用了两种机制可解释性工具：LogitLens (Belrose et al., 2023; nostalgebraist, 2020) 和中心核对齐分析 (CKA) (Davari et al., 2023; Kornblith et al., 2019)。

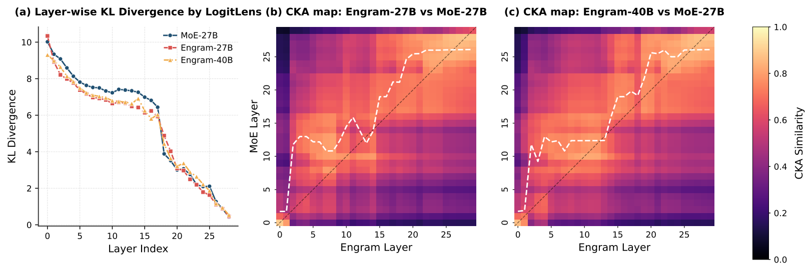

Figure 4 | **Analysis of representational alignment and convergence speed.** (a) Layer-wise KL Divergence via LogitLens (nostalgebraist, 2020). The consistently lower divergence in early layers indicates that Engram accelerates prediction convergence. (b-c) Similarity heatmap computed by CKA (Kornblith et al., 2019). The distinct upward shift of the high-similarity diagonal demonstrates that =Engram's shallow layers are functionally equivalent to deeper layers of the MoE model, effectively increasing the model's depth.

6. Analysis

In this section, we investigate the internal mechanisms of Engram, including its effective depth (Section 6.1), core module design (Section 6.2), and parametric sensitivity (Section 6.3). Additionally, we evaluate the inference throughput with offloading (Section 6.4) and conclude with a case study (Section 6.5).

6.1. Is Engram functionally equivalent to increasing the model's depth?

Current LLMs lack a dedicated knowledge lookup primitive and they rely on computation to simulate memory recall. As shown in Table 3, to recognize the entity `"Diana, Princess of Wales"`, an LLM must consume multiple layers of Attention and FFNs to progressively compose features (Ghandeharioun et al., 2024; Jin et al., 2025; Li and Subramani, 2025), a process that could theoretically be identified via a knowledge lookup operation.

Given this, we posit that by equipping the model with an explicit knowledge lookup capability, Engram effectively mimics an increase in model depth by relieving the model of the early stages of feature composition. To validate this hypothesis, we employ two mechanistic interpretability tools: LogitLens (Belrose et al., 2023; nostalgebraist, 2020) and Centered Kernel Alignment analysis (CKA) (Davari et al., 2023; Kornblith et al., 2019).

6.1.1. Accelerated Prediction Convergence

We first analyze the evolution of predictions across layers using LogitLens (nostalgebraist, 2020). By projecting each intermediate layer's hidden state with the final LM Head, we compute the Kullback–Leibler divergence (Kullback and Leibler, 1951) between the intermediate output distribution and the model's final output distribution. This metric quantifies how close a latent representation is to being "prediction-ready" (Belrose et al., 2023; Csordás et al., 2025).

Figure 4 (a) reports the layer-wise KL divergence. Compared to the MoE baseline, both Engram variants exhibit systematically smaller KL divergence, with the most pronounced gap

Table 3 | **Entity resolution example reproduced from** Ghandeharioun et al. (2024)。本表格展示了 LLM 如何通过注意力层和前馈网络层逐步整合上下文标记，以构建实体 `` ``Diana, Princess of Wales'' ``的内部表征。"潜在状态翻译"列显示了通过 PatchScope (Ghandeharioun et al., 2024) 为最后一个标记 `` ``Wales'' `` 自动生成的文本，而"解释"列则呈现了原作者提供的手动解释。

Layer	Latent State Translation	Explanation
1-2	: Country in the United Kingdom	**Wales**
3	: Country in Europe	**Wales**
4	: Title held by female sovereigns in their own right or by queens consort	**Princess of Wales** (unspecific)
5	: Title given to the wife of the Prince of Wales (and later King)	**Princess of Wales** (unspecific)
6	: Diana, Princess of Wales (1961-1997), the first wife of Prince Charles, Prince of Wales, who was famous for her beauty and humanitarian work	**Diana, Princess of Wales**

Table 3 | **Entity resolution example reproduced from** Ghandeharioun et al. (2024). This table illustrates how LLMs gradually integrate context tokens through layers of attention and FFNs to construct the internal representation of the entity: `"Diana, Princess of Wales"`. The "Latent State Translation" column displays the automatically generated text for the last token: `"Wales"` by PatchScope (Ghandeharioun et al., 2024), while the "Explanation" column presents the manual interpretation provided by the original authors.

Layer	Latent State Translation	Explanation
1-2	: Country in the United Kingdom	**Wales**
3	: Country in Europe	**Wales**
4	: Title held by female sovereigns in their own right or by queens consort	**Princess of Wales** (unspecific)
5	: Title given to the wife of the Prince of Wales (and later King)	**Princess of Wales** (unspecific)
6	: Diana, Princess of Wales (1961-1997), the first wife of Prince Charles, Prince of Wales, who was famous for her beauty and humanitarian work	**Diana, Princess of Wales**

appearing in the early blocks. The steeper descent in the Engram curves indicates that the model finishes feature composition much faster. This observation aligns with our hypothesis: by accessing external knowledge explicitly, Engram reduces the computational steps required, thereby reaching high-confidence, valid predictions earlier in the network hierarchy.

6.1.2. Representational Alignment and Effective Depth

To further investigate whether Engram layers semantically correspond to deeper layers of the baseline, we employ Centered Kernel Alignment (CKA), a widely established metric for comparing representational structures (Kornblith et al., 2019; Kriegeskorte et al., 2008). Given two sets of representations X and Y (e.g., activations from different models or layers), CKA is defined as:

$$\mathrm{CKA}(K,L) = \frac{\mathrm{HSIC}(K,L)}{\sqrt{\mathrm{HSIC}(K,K)\mathrm{HSIC}(L,L)}} \tag{8}$$

where $K = XX^\top$ and $L = YY^\top$ denote the Gram matrices (using a linear kernel) and HSIC is Hilbert-Schmidt Independence Criterion (Gretton et al., 2005). We employ a minibatch implementation with an unbiased estimator of HSIC (Davari et al., 2023) and evaluate on the Few-NERD dataset (Ding et al., 2021), extracting hidden states corresponding to the final token of named entities.

To rigorously quantify the layer-wise correspondence, we first compute the pairwise CKA similarity matrix $S \in [0,1]^{L\times L}$, where L is the number of layers. We then introduce a soft alignment index a_j, defined as the weighted centroid of the top-k most similar MoE layers for each Engram layer j:

$$a_j = \frac{\sum_{i\in\mathcal{I}_j} S_{i,j}\cdot i}{\sum_{i\in\mathcal{I}_j} S_{i,j}}, \quad \text{where } \mathcal{I}_j = \underset{i}{\mathrm{argtop}k}(S_{i,j}). \tag{9}$$

Here, $S_{i,j}$ denotes the similarity score between MoE layer i and Engram layer j. The index a_j

5.1.1. Accelerated Prediction Convergence

我们首先使用 LogitLens (nostalgebraist, 2020) 分析预测在不同层间的演变。通过将每个中间层的隐藏状态用最终的 LM Head 进行投影，我们计算中间输出分布与模型最终输出分布之间的 Kullback–Leibler 散度 (Kullback and Leibler, 1951)。该指标量化了潜在表示距离"准备好进行预测"(Belrose et al., 2023; Csordás et al., 2025) 的程度。

Figure 3 (a) 报告了逐层的 KL 散度。与 MoE 基线相比，两种 Engram 变体都表现出系统性的更小 KL 散度，最显著的差距出现在早期块中。Engram 曲线更陡峭的下降表明模型完成特征组合的速度要快得多。这一观察结果与我们的假设一致：通过显式访问外部知识，Engram 减少了所需的计算步骤，从而在网络层级中更早地达到高置信度的有效预测。

5.1.2. Representational Alignment and Effective Depth

为了进一步研究记忆印迹层是否在语义上对应于基线的更深层，我们采用了中心核对齐 (CKA)，这是一种广泛用于比较表示结构的度量标准 (Kornblith et al., 2019; Kriegeskorte et al., 2008)。给定两组表示 X 和 Y（例如，来自不同模型或层的激活），CKA 定义为：

$$\mathrm{CKA}(K,L) = \frac{\mathrm{HSIC}(K,L)}{\sqrt{\mathrm{HSIC}(K,K)\mathrm{HSIC}(L,L)}} \tag{7}$$

其中 $K = XX^\top$ 和 $L = YY^\top$ 表示 Gram 矩阵（使用线性核），HSIC 是希尔伯特-施密特独立性准则 (Gretton et al., 2005)。我们采用带有 HSIC 无偏估计器的小批量实现 (Davari et al., 2023)，并在 Few-NERD 数据集 (Ding et al., 2021) 上进行评估，提取命名实体最后一个标记对应的隐藏状态。

为了严格量化层间对应关系，我们首先计算成对 CKA 相似度矩阵 $S \in [0,1]^{L\times L}$，其中 L 是层数。然后我们引入一个软对齐指数 a_j，定义为每个记忆印迹层 j 对应的前 k 个最相似的 MoE 层的加权质心：

$$a_j = \frac{\sum_{i\in\mathcal{I}_j} S_{i,j}\cdot i}{\sum_{i\in\mathcal{I}_j} S_{i,j}}, \quad \text{where } \mathcal{I}_j = \underset{i}{\mathrm{argtop}k}(S_{i,j}). \tag{8}$$

这里，$S_{i,j}$ 表示 MoE 层 i 与记忆印迹层 j 之间的相似度分数。指数 a_j 作为记忆印迹层 j 对应的"有效 MoE 深度"的稳健代理，利用前 k 过滤（取 $k = 5$）来减轻低相似度噪声。

Figure 3 (b)–(c) 可视化了叠加有软对齐曲线（白色虚线）的相似度热图。我们观察到一个明显的向上偏离对角线的偏移，这意味着 $a_j > j$ 对于广泛的层范围成立。例如，Engram-27B 第 5 层形成的表示与 MoE 基线大约第 12 层的表示最为接近。

这种一致的偏离对角线偏移，与 LogitLens 结果 (节 5.1.1) 相符，证实了记忆印迹在更早的层实现了更深的表示。这验证了我们的核心假设：通过显式查找绕过早期特征组合，记忆印迹在功能上等同于增加了模型的有效深度。

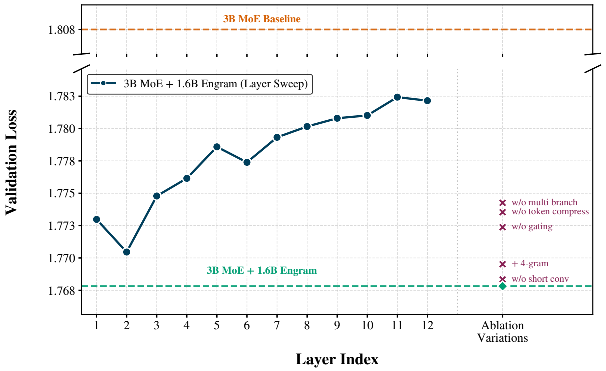

Figure 5 | **Architecture ablation results.** We compare the 3B MoE baseline against Engram variations in two settings: (1) Layer Sensitivity (dark blue curve): Sweeping the insertion depth of a single Engram module confirms that early injection (Layer 2) is optimal, whereas efficacy degrades in deeper layers. (2) Component Ablation (Right Markers): Removing sub-modules from the reference configuration demonstrates the importance of multi-branch integration, tokenizer compression, and context-aware gating.

serves as a robust proxy for the "effective MoE depth" corresponding to Engram layer j, utilizing top-k filtering (with $k = 5$) to mitigate low-similarity noise.

Figure 4 (b)–(c) visualize the similarity heatmaps overlayed with the soft alignment curve (dashed white line). We observe a distinct upward shift from the diagonal, meaning that $a_j > j$ for a wide range of layers. For instance, the representations formed at layer 5 of Engram-27B align most closely with those at approximately layer 12 of the MoE baseline.

The consistent off-diagonal shift, which aligns with the LogitLens results (Section 6.1.1), confirms that Engram achieves deeper representations at earlier layers. This validates our central hypothesis: by bypassing early-stage feature composition via explicit lookups, Engram is functionally equivalent to increasing the model's effective depth.

6.2. Structural Ablation and Layer Sensitivity

In this section, we ablate Engram under a controlled setting to investigate the effectiveness of each key module design. Unless otherwise specified, the backbone is a 12-layer 3B MoE model (0.56B activated parameters) trained for 100B tokens. Figure 5 reports validation loss. The dashed orange line denotes the 3B MoE baseline (Val Loss = 1.808).

Reference configuration. We augment the backbone with a fixed 1.6B-parameter Engram memory. Our reference model uses {2, 3}-grams and inserts Engram at Layers 2 and 6, achieving Val Loss = 1.768, a substantial improvement over the MoE baseline ($\Delta = 0.04$). All structural ablations below are defined relative to this reference.

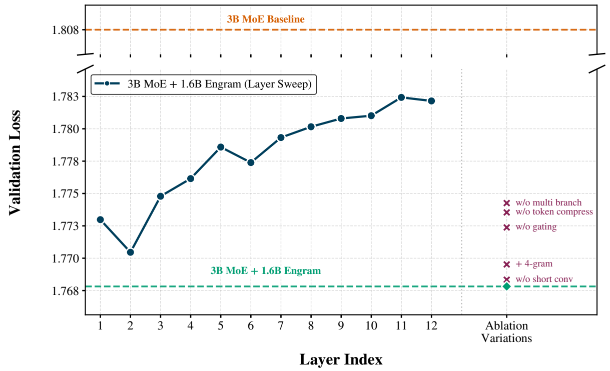

Figure 4 | **Architecture ablation results.** 我们在两种设置下比较 3B MoE 基线与记忆印迹变体：(1) 层敏感性（深蓝色曲线）：扫描单个记忆印迹模块的插入深度证实，早期注入（第 2 层）是最优的，而在更深层注入效果会下降。(2) 组件消融（右侧标记）：从参考配置中移除子模块，证明了多分支集成、分词器压缩和上下文感知门控的重要性。

5.2. Structural Ablation and Layer Sensitivity

在本节中，我们在受控设置下对 Engram 进行消融研究，以探究每个关键模块设计的有效性。除非另有说明，主干网络是一个 12 层、30 亿参数的 MoE 模型（激活参数为 5.6 亿），训练了 1000 亿个词元。Figure 4 报告了验证损失。橙色虚线表示 30 亿参数 MoE 基线（验证损失 = 1.808）。

Reference configuration. 我们使用一个固定的 16 亿参数 Engram 记忆库来增强主干网络。我们的参考模型使用 {2, 3}-grams 并在第 2 层和第 6 层插入 Engram，实现了验证损失 = 1.768，相较于 MoE 基线（$\Delta = 0.04$）有显著提升。以下所有结构消融均相对于此参考模型定义。

Where should memory be injected? 为了研究深度敏感性，我们保持 Engram 预算固定（16 亿），但将其合并为单个 Engram 模块，并将其插入层从 1 到 12 进行扫描（Figure 4 中的深蓝色"层扫描"曲线）。该实验揭示了 Engram 放置位置固有的权衡。

A placement trade-off. 早期注入 Engram 使其能够在主干网络消耗计算深度之前卸载局部模式重建，这与主干网络的自然分层处理方式一致 (Ghandeharioun et al., 2024; Jin et al., 2025; Li and Subramani, 2025; Tenney et al., 2019)。然而，这会带来门控精度的代价：早期的隐藏状态尚未通过注意力聚合足够的全局上下文，并且并行分支缺乏细粒度调制所需的表征差异 (Xie et al., 2025; Zhu et al., 2025)。因此，最优放置需要平衡 (i) 早期卸载静态局部模式和 (ii) 后期利用更强的上下文查询进行门控。

扫描结果显示，第 2 层实现了最佳的单层性能（验证损失 = 1.770），优于第 1 层，并且随

Where should memory be injected? To study depth sensitivity, we keep the Engram budget fixed (1.6B) but consolidate it into a single Engram module, and sweep its insertion layer from 1 to 12 (dark blue "Layer Sweep" curve in Figure 5). This experiment exposes an inherent trade-off in Engram placement.

A placement trade-off. Injecting Engram early allows it to offload local pattern reconstruction before the backbone expends computational depth, aligning with the backbone's natural hierarchical processing (Ghandeharioun et al., 2024; Jin et al., 2025; Li and Subramani, 2025; Tenney et al., 2019). However, this incurs a cost in gating precision: early hidden states have not yet aggregated sufficient global context via attention, and the parallel branches lack the representational divergence required for fine-grained modulation (Xie et al., 2025; Zhu et al., 2025). Consequently, optimal placement requires balancing (i) offloading static local patterns early and (ii) utilizing stronger contextual queries for gating later.

The sweep shows that Layer 2 achieves the best single-layer performance (Val Loss = 1.770), outperforming Layer 1 and degrading as the insertion point moves deeper. This indicates that one round of attention is already sufficient to provide a meaningfully contextualized $\mathbf{h}_t$ for gating, while still being early enough to replace the backbone's bottom-layer local aggregation.

While Layer 2 is optimal under a single injection constraint, we find that dividing the same 1.6B memory into two smaller modules (achieved by reducing the embedding dimension d_{mem}) and placing them at Layers 2 and 6 performs even better (Val Loss = 1.768). This layered design reconciles the trade-off by combining early intervention with rich, late-stage contextual gating. More importantly, layered insertion also provides a practical system advantage, enabling better utilization of the memory hierarchy as discussed in Section 2.5.

Which components matter? Starting from the reference configuration, we ablate individual design choices while keeping the Engram parameter budget fixed. Results are denoted by markers in Figure 5. We find that three components yield the most significant gains: (i) branch-specific fusion within the multi-branch backbone, (ii) context-aware gating, and (iii) tokenizer compression. Removing any of these causes the largest regressions in validation loss. Specifically, for the "w/o multi branch" ablation, we retain the mHC backbone structure but replace the branch-specific gating with a single Engram fusion applied to the hidden states after the pre-mapping $\mathcal{H}^{pre}$ (Xie et al., 2025).

Other changes have smaller effects: removing the lightweight depthwise convolution only marginally degrades performance. Allocating capacity to 4-grams is slightly suboptimal under a fixed 1.6B budget—likely because it dilutes capacity from the more frequent 2/3-gram patterns—though we do not rule out that higher-order N-grams become beneficial at larger memory scales.

6.3. Sensitivity Analysis

To characterize the functional contribution of the Engram module, we evaluate the model by completely suppressing the sparse embedding output during inference while keeping the backbone unchanged. Crucially, this post-hoc ablation induces a **training–inference inconsistency**, potentially introducing noise in complex, mixed-capability tasks. Consequently, we prioritize the analysis of *Factual Knowledge* and *Reading Comprehension*—the two extremes of the sensitivity spectrum—which exhibit the highest signal-to-noise ratio under this stress test.

As shown in Figure 6, the results reveal a sharp functional dichotomy. Factual knowledge

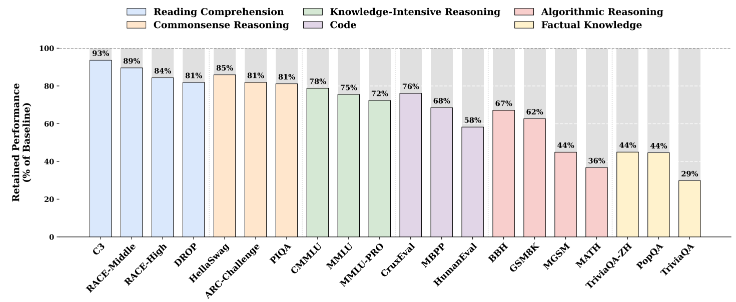

Figure 5 | **Retained performance under Engram ablation.** 事实性知识高度依赖于记忆印迹模块，而阅读理解能力则主要由骨干网络保留。

着插入点变深而性能下降。这表明，一轮注意力已经足以为门控提供一个有意义的、上下文化的 $\mathbf{h}_t$，同时仍然足够早，可以替代主干网络底层的局部聚合。

虽然第 2 层在单次注入约束下是最优的，但我们发现，将相同的 16 亿记忆库分成两个较小的模块（通过减少嵌入维度 d_{mem} 实现）并将其放置在第 2 层和第 6 层，性能甚至更好（验证损失 = 1.768）。这种分层设计通过结合早期干预和丰富的后期上下文门控，调和了上述权衡。更重要的是，分层插入还提供了实际的系统优势，能够更好地利用内存层次结构，如 节 2.5 中所述。

Which components matter? 从参考配置出发，我们消融了各个设计选择，同时保持 Engram 参数预算固定。结果由 Figure 4 中的标记点表示。我们发现三个组件带来了最显著的收益：(i) 多分支主干网络内的分支特定融合，(ii) 上下文感知门控，以及 (iii) 分词器压缩。移除其中任何一个都会导致验证损失出现最大的回归。具体来说，对于"无多分支"消融，我们保留了 mHC 主干结构，但将分支特定门控替换为应用于预映射 $\mathcal{H}^{pre}$ (Xie et al., 2025) 后隐藏状态的单一 Engram 融合。其他改动的影响较小：移除轻量级深度卷积仅会轻微降低性能。在固定的 16 亿参数预算下，将容量分配给 4-gram 略有不足——可能是因为它稀释了来自更频繁的 2/3-gram 模式的容量——尽管我们不排除更高阶的 N-gram 在更大的内存规模下会变得有益。

5.3. Sensitivity Analysis

为了表征记忆印迹模块的功能贡献，我们在推理过程中完全抑制稀疏嵌入输出，同时保持骨干网络不变，以此来评估模型。至关重要的是，这种事后消融会引发 **training–inference inconsistency**，可能在复杂的混合能力任务中引入噪声。因此，我们优先分析 *Factual Knowledge* 和 *Reading Comprehension*——这两个位于敏感性谱两端的极端情况——在此压力测试下表现出最高的信噪比。

如Figure 5所示，结果揭示了一种鲜明的功能二分法。事实性知识基准测试遭受了灾难性的

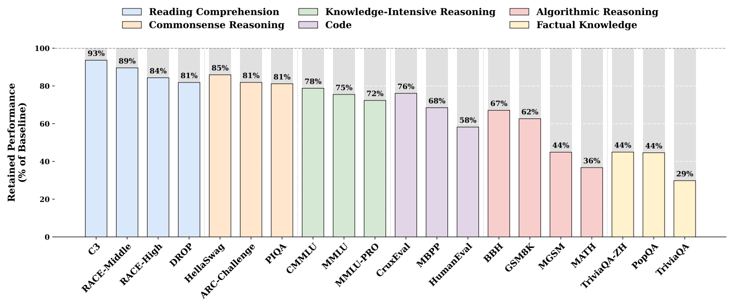

Figure 6 | **Retained performance under Engram ablation.** Factual knowledge relies heavily on the Engram module, whereas reading comprehension is largely preserved by the backbone.

benchmarks suffer a catastrophic collapse, retaining only 29–44% of the original performance (e.g., TriviaQA at 29%), confirming that the Engram module acts as the primary repository for parametric knowledge. Conversely, reading comprehension tasks are remarkably resilient, retaining 81–93% (e.g., C3 at 93%), suggesting that context-grounded tasks rely primarily on the backbone's attention mechanism rather than Engram.

6.4. System Efficiency

A pivotal system advantage of Engram over routing-based MoE is that its sparse activations are addressed by explicit, static hash IDs. This yields a strictly deterministic memory access pattern: indices for the next Engram lookup are fixed once the token sequence is known and can be computed before the corresponding layer executes.

Experimental Setup. We implemented an inference harness based on nano-vLLM[1]—a streamlined prototype of the industry-standard vLLM engine (Kwon et al., 2023). To obtain a clean latency baseline without the confounding communication patterns of Expert Parallel in MoE, we benchmark on two dense backbones (Dense-4B and Dense-8B). We insert a massive 100B-parameter Engram layer into the second Transformer block, with the entire embedding table resident in host DRAM. During inference, the system prefetches embeddings for the Engram layer asynchronously, overlapping the PCIe transfer with the computation of the first block.

Results. As detailed in Table 4, offloading a 100B-parameter embedding table incurs a negligible throughput penalty, peaking at only 2.8% on the 8B backbone. This confirms that the compute intensity of early dense blocks provides a sufficient temporal window to mask the retrieval latency. Crucially, the effective communication volume per step scales with the number of activated slots rather than the total embedding table size.

Crucially, this experiment serves as a conservative baseline. While the hierarchical design in Section 2.5 exploits Zipfian locality to cache frequent items in HBM, our experimental setup forces *all* retrievals to traverse the PCIe bus from host memory. The fact that this baseline

[1]https://github.com/GeeeekExplorer/nano-vllm

崩溃，仅保留了原始性能的 29–44%（例如，TriviaQA 为 29%），这证实了记忆印迹模块是参数化知识的主要存储库。相反，阅读理解任务表现出显著的韧性，保留了 81–93%（例如，C3 为 93%），这表明基于上下文的任务主要依赖于骨干网络的注意力机制，而非记忆印迹。

5.4. System Efficiency

Engram 相对于基于路由的 MoE 的一个关键系统优势在于，其稀疏激活是通过显式、静态的哈希 ID 进行寻址的。这产生了一种严格确定性的内存访问模式：一旦已知令牌序列，下一次 Engram 查找的索引就是固定的，并且可以在相应层执行之前计算出来。

Experimental Setup. 我们基于 nano-vLLM[1]——一个行业标准 vLLM 引擎的简化原型 (Kwon et al., 2023)——实现了一个推理测试框架。为了获得一个不受 MoE 中专家并行通信模式干扰的纯净延迟基线，我们在两个稠密骨干网络（Dense-4B 和 Dense-8B）上进行基准测试。我们在第二个 Transformer 块中插入了一个庞大的 100B 参数 Engram 层，整个嵌入表驻留在主机 DRAM 中。在推理过程中，系统异步预取 Engram 层的嵌入，使 PCIe 传输与第一个块的计算重叠。

Table 4 | **End-to-end Inference Throughput**。我们测量了将一个 100B 参数的 Engram 层完全卸载到主机内存时的推理吞吐量。

Experimental Setup		
Hardware		NVIDIA H800
Workload		512 Sequences
Sequence Length		Uniform(100, 1024)
Throughput Results		
Base Model	**Configuration**	**Throughput (tok/s)**
4B-Dense	Baseline	9,031.62
	+ 100B Engram (CPU Offload)	8,858.28
8B-Dense	Baseline	6,315.52
	+ 100B Engram (CPU Offload)	6,140.02

Results. 如表 4中详述，卸载一个 100B 参数的嵌入表带来的吞吐量惩罚可以忽略不计，在 8B 骨干网络上峰值仅为 2.8%。这证实了早期稠密块的计算强度提供了一个足够的时间窗口来掩盖检索延迟。至关重要的是，每一步的有效通信量随激活槽的数量缩放，而不是随整个嵌入表的大小缩放。

[1]https://github.com/GeeeekExplorer/nano-vllm

Table 4 | **End-to-end Inference Throughput**. We measure infernece throughput with a 100B-parameter Engram layer entirely offloaded to host memory.

Experimental Setup		
Hardware		NVIDIA H800
Workload		512 Sequences
Sequence Length		Uniform(100, 1024)
Throughput Results		
Base Model	**Configuration**	**Throughput (tok/s)**
4B-Dense	Baseline	9,031.62
	+ 100B Engram (CPU Offload)	8,858.28
8B-Dense	Baseline	6,315.52
	+ 100B Engram (CPU Offload)	6,140.02

retrieval strategy yields minimal overhead strongly suggests that a fully optimized, locality-aware implementation would incur negligible throughput penalty.

6.5. Case Study: Gating Visualization

In Section 2.3, we introduced the context-aware gating mechanism, designed to dynamically modulate the integration of retrieved static memory into the backbone. To empirically validate whether Engram behaves as intended, we visualize the gating scalar α_t of Engram-27B[2] across various samples in Figure 7.

The results demonstrate a distinct pattern of selectivity. The gating mechanism consistently activates (shown in red) upon completing local, static patterns. In English, we observe strong activations on multi-token named entities (e.g., "Alexander the Great", "the Milky Way") and formulaic phrases (e.g., "By the way", "Princess of Wales"). This behavior generalizes effectively across languages. In the Chinese examples, Engram identifies and retrieves distinct idiomatic expressions and historical entities, such as "Four Great Inventions" (四大发明) and "Zhang Zhongjing" (张仲景). These qualitative results confirm that Engram successfully identifies and handles stereotyped linguistic dependencies, effectively relieving the Transformer backbone from memorizing these static associations.

7. Related Work

***N*-gram Modeling and Embedding Scaling.** Originating from Shannon's framework (Shannon, 1948), N-gram models rely on local history to predict tokens, traditionally employing smoothing techniques (Katz, 1987; Kneser and Ney, 1995) to mitigate data sparsity. Despite the paradigm shift toward neural architectures (Bengio et al., 2003) for capturing long-range dependencies, the computational efficiency of N-gram lookups has been preserved in modern representation learning, as exemplified by seminal works like FastText (Bojanowski et al., 2017).

[2]As detailed in our architecture setup, this model utilizes a mHC ($M = 4$) with Engram modules inserted at layers 2 and 15. Consequently, for any given token, the model computes a total of 8 distinct gating scalars. We observe that not every branch encodes interpretable activation patterns. For the clarity of this visualization, we select and display the gating values most strongly correlated with semantic pattern matching.

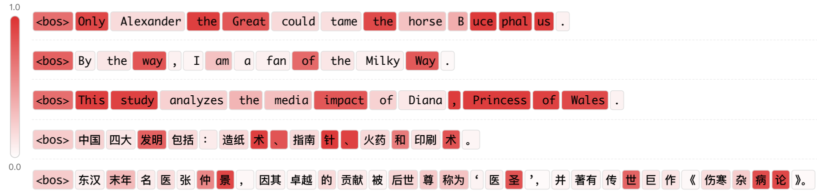

Figure 6 | **Visualization of the gating mechanism of Engram.** 热图强度对应于门控标量 $\alpha_t \in [0,1]$ 的大小，其中更深的红色表示更强的激活。因为 Engram 作用于后缀 N-gram（此处为 $N = 3$），在特定标记 x_t 上的高激活意味着，以该标记为结尾的前置标记（例如，在 t 处结束的短语）被识别为从记忆中有效检索到的静态模式。

至关重要的是，这个实验提供了一个保守的基线。虽然节 2.5中的分层设计利用 Zipfian 局部性将频繁项缓存在 HBM 中，但我们的实验设置强制所有 *all* 检索都必须通过 PCIe 总线从主机内存获取。这种基线检索策略产生极小的开销这一事实，强烈表明一个完全优化的、具备局部性感知能力的实现将带来可忽略的吞吐量惩罚。

5.5. Case Study: Gating Visualization

在节 2.3中，我们介绍了上下文感知的门控机制，旨在动态调节从主干网络检索到的静态记忆的整合。为了通过实证验证 Engram 是否按预期运行，我们可视化了 Engram-27B[2]在不同样本中的门控标量 α_t（见Figure 6）。

结果展示了一种明显的选择性模式。该门控机制在完成局部的、静态的模式时持续激活（以红色显示）。在英语中，我们观察到在多标记命名实体（例如，"Alexander the Great"、"the Milky Way"）和公式化短语（例如，"By the way"、"Princess of Wales"）上有强烈的激活。这种行为在不同语言中都能有效泛化。在中文示例中，Engram 识别并检索了独特的习语表达和历史实体，例如" 四大发明"（**四大发明**）和" 张仲景"（**张仲景**）。这些定性结果证实，Engram 成功地识别并处理了刻板化的语言依赖关系，有效地减轻了 Transformer 主干网络记忆这些静态关联的负担。

6. Related Work

***N*-gram Modeling and Embedding Scaling.** 源于香农的框架 (Shannon, 1948)，N-元模型依赖局部历史来预测词元，传统上采用平滑技术 (Katz, 1987; Kneser and Ney, 1995) 来缓解数据稀疏性问题。尽管范式已转向神经架构 (Bengio et al., 2003) 以捕捉长程依赖关系，但 N-元查找的计算效率在现代表示学习中得以保留，例如 FastText (Bojanowski et al., 2017) 等开创性工

[2]正如我们的架构设置中所详述，该模型使用了一个 mHC（$M = 4$），并在第 2 层和第 15 层插入了 Engram 模块。因此，对于任何给定的标记，模型会计算总共 8 个不同的门控标量。我们观察到并非每个分支都编码可解释的激活模式。为了本可视化的清晰性，我们选择并展示了与语义模式匹配最强烈相关的门控值。

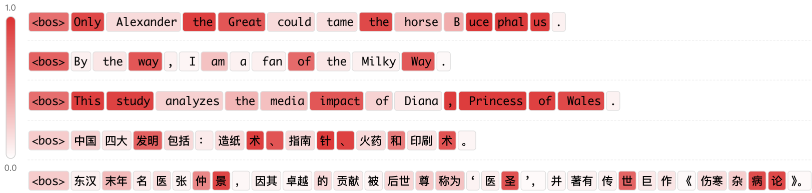

Figure 7 | **Visualization of the gating mechanism of Engram.** The heatmap intensity corresponds to the magnitude of the gating scalar $\alpha_t \in [0, 1]$, where darker red indicates stronger activation. Because Engram operates on suffix N-grams (here $N = 3$), a high activation on a specific token x_t implies that the preceding tokens culminating in that token (e.g., the phrase ending at t) are recognized as a static pattern effectively retrieved from memory.

Recently, this paradigm has resurged as *embedding scaling*. While architectures such as Per-Layer Embeddings (Team, 2025) and DeepEmbed (RWKV Team, 2025) expand capacity via massive tables, a distinct line of pioneering research—most relevant to our approach—integrates compositional N-gram structures directly into the representation space. SuperBPE (Liu et al., 2025) and SCONE (Yu et al., 2025) explicitly target high-frequency patterns: the former by merging multi-word expressions into "superword" tokens, and the latter via an auxiliary encoding model. In parallel, OverEncoding (Huang et al., 2025a) and Byte Latent Transformer (BLT) (Pagnoni et al., 2025) adopt hash N-gram embeddings to capture local dependencies at the token and byte levels, respectively. These studies collectively demonstrate the efficacy of scaling parameters through N-gram representations with minimal computational overhead. While these approaches offer significant gains in their respective settings, our work diverges fundamentally in two key dimensions.

- First, regarding modeling and evaluation protocols. Prior approaches often treat N-gram embeddings as external augmentations without validating their efficiency under strictly fair comparison protocols. For instance, SCONE (Yu et al., 2025) is inference-focused and relies on auxiliary modules that incur additional training FLOPs. Similarly, OverEncoding (Huang et al., 2025a) fails to yield meaningful improvements on sparse MoE backbones even under a non-isoparametric setting. In contrast, we treat conditional memory as a first-class modeling primitive instantiated via the carefully designed Engram module. By rigorously evaluating this design within our *Sparsity Allocation* framework, we demonstrate its clear advantage over strictly iso-parameter and iso-FLOPs MoE baselines.
- Second, from a system perspective, we advocate for algorithm-system co-design. Existing approaches place embeddings strictly at the input layer (Layer 0), which inherently serializes memory access and computation (Huang et al., 2025a; Yu et al., 2025). Engram, conversely, strategically injects memory into deeper layers to enable communication-computation overlap. Furthermore, by exploiting the inherent Zipfian distribution of N-grams, we could maximize the utility of the hardware memory hierarchy. This holistic design allows Engram to scale to massive parameters with negligible inference overhead.

Mixture-of-Experts. MoE architectures decouple model capacity from computational cost by conditionally activating a sparse subset of experts per token, a paradigm introduced by Shazeer et al. (2017). Subsequent innovations such as GShard (Lepikhin et al., 2020), BASE (Lewis et al., 2021), Switch Transformer (Fedus et al., 2022) and GLaM (Du et al., 2022) enabled super-

作所示。

最近,这一范式以 *embedding scaling* 的形式重新兴起。虽然诸如 Per-Layer Embeddings (Team, 2025) 和 DeepEmbed (RWKV Team, 2025) 等架构通过庞大的表格扩展了容量,但另一条开创性的研究路线——与我们的方法最相关——将组合式的 N-元结构直接整合到表示空间中。SuperBPE (Liu et al., 2025) 和 SCONE (Yu et al., 2025) 明确针对高频模式:前者通过将多词表达式合并为"超词"词元,后者则通过辅助编码模型实现。与此同时,OverEncoding (Huang et al., 2025a) 和字节潜在变换器(BLT)(Pagnoni et al., 2025) 分别采用哈希 N-元嵌入来捕捉词元级和字节级的局部依赖关系。这些研究共同证明了通过 N-元表示扩展参数的有效性,且计算开销极小。虽然这些方法在各自的情境下带来了显著收益,但我们的工作在两个关键维度上存在根本性差异。

- 首先,关于建模和评估协议。先前的方法通常将 N-元嵌入视为外部增强,而未在严格公平的比较协议下验证其效率。例如,SCONE (Yu et al., 2025) 侧重于推理,并依赖会产生额外训练 FLOPs 的辅助模块。类似地,即使在非等参数设置下,OverEncoding (Huang et al., 2025a) 在稀疏 MoE 骨干网络上也无法产生有意义的改进。相比之下,我们将条件记忆视为通过精心设计的 Engram 模块实例化的一等建模原语。通过在 *Sparsity Allocation* 框架内严格评估此设计,我们证明了其相对于严格等参数和等 FLOPs 的 MoE 基线的明显优势。
- 其次,从系统视角出发,我们倡导算法-系统协同设计。现有方法将嵌入严格置于输入层(第 0 层),这本质上串行化了内存访问和计算 (Huang et al., 2025a; Yu et al., 2025)。相反,Engram 策略性地将内存注入到更深层,以实现通信与计算的重叠。此外,通过利用 N-元固有的齐夫分布特性,我们可以最大化硬件内存层次结构的效用。这种整体设计使得 Engram 能够扩展到海量参数,同时推理开销可忽略不计。

Mixture-of-Experts. MoE 架构通过为每个 token 有条件地激活一个稀疏的专家子集,将模型容量与计算成本解耦,这一范式由 Shazeer et al. (2017) 引入。后续的创新,如 GShard (Lepikhin et al., 2020)、BASE (Lewis et al., 2021)、Switch Transformer (Fedus et al., 2022) 和 GLaM (Du et al., 2022),实现了超线性的参数扩展,同时保持了恒定的推理成本。最近,DeepSeek-MoE (Dai et al., 2024) 展示了卓越的效率,通过细粒度的专家分割和共享专家隔离,在同等激活参数下显著优于稠密模型。采用这种架构,最先进的模型如 DeepSeek-V3 (Liu et al., 2024a) 和 Kimi-k2 (Team et al., 2025) 进一步将总参数量推至数千亿规模。

Memory Network. 关于记忆增强网络的研究旨在扩展模型容量而不成比例地增加计算成本,大致可分为参数化和非参数化方法。参数化记忆方法,如 PKM (Lample et al., 2019)、PEER (He, 2024)、Selfmem (Cheng et al., 2023b)、Memory+ (Berges et al., 2025) 和 UltraMem (Huang et al., 2025b,c),将大规模、稀疏的键值存储直接集成到模型层中,从而显著增加容量,而对 FLOPs 的影响微乎其微。相反,非参数化记忆方法如 REALM (Guu et al., 2020)、RETRO (Borgeaud et al., 2022; Wang et al., 2023) 和 PlugLM (Cheng et al., 2023a),将知识存储与模型处理解耦,

linear parameter scaling while maintaining constant inference costs. More recently, DeepSeek-MoE (Dai et al., 2024) demonstrated superior efficiency, significantly outperforming dense models with equivalent active parameters via fine-grained expert segmentation and shared expert isolation. Adopting this architecture, state-of-the-art models such as DeepSeek-V3 (Liu et al., 2024a) and Kimi-k2 (Team et al., 2025) have further pushed total parameters to hundreds of billions scale.

Memory Network. Research on memory-augmented networks aims to expand model capacity without a proportional increase in computational cost, broadly categorized into parametric and non-parametric approaches. Parametric memory methods, such as PKM (Lample et al., 2019), PEER (He, 2024), Selfmem (Cheng et al., 2023b), Memory+ (Berges et al., 2025) and Ultra-Mem (Huang et al., 2025b,c), integrate large-scale, sparse key-value stores directly into the model layers, thereby significantly increasing capacity with negligible impact on FLOPs. Conversely, non-parametric memory approaches like REALM (Guu et al., 2020), RETRO (Borgeaud et al., 2022; Wang et al., 2023), and PlugLM (Cheng et al., 2023a) decouple knowledge storage from model processing, treating the external memory as an editable and scalable key-value store that allows the model to adapt to evolving information without retraining.

Mechanisms of Knowledge Storage. Parallel to capacity scaling, substantial research has scrutinized the internal mechanisms governing how Transformers encode and retrieve factual knowledge. The Feed-Forward Networks (FFNs) are widely hypothesized to function as Key-Value memories (Geva et al., 2021). Under this framework, the first layer acts as a pattern detector ("keys") while the second layer projects specific information into the residual stream ("values"). This modularity is evidenced by the identification of specific “knowledge neurons” responsible for storing distinct facts (Dai et al., 2022). Further validation is provided by causal tracing methodologies, which map the information flow of factual recall to specific FFN layers (Meng et al., 2022). These insights have enabled precise model editing algorithms such as ROME (Meng et al., 2022) and MEMIT (Meng et al., 2023), which allow for the direct update of factual associations without retraining. Moreover, investigations into internal representations, such as those in Othello-GPT (Li et al., 2023a), suggest that these storage mechanisms may facilitate the emergence of structured “world models” rather than mere statistical memorization.

8. Conclusion

In this work, we introduce **conditional memory** as a complementary sparsity axis to the prevailing conditional computation paradigm (MoE), aiming to resolve the inefficiency of simulating knowledge retrieval through dynamic computation. We instantiate this concept via Engram, a module that modernizes classic N-gram embeddings to enable scalable, constant-time $O(1)$ lookups for static patterns

By formulating the *Sparsity Allocation* problem, we uncover a U-shaped scaling law, demonstrating that a hybrid allocation of sparse capacity between MoE experts and Engram memory strictly outperforms pure MoE baselines. Guided by this law, we scale Engram to 27B parameters, achieving superior performance across diverse domains. Notably, while the memory module intuitively aids knowledge retrieval, we observe even larger gains in general reasoning, code, and mathematics.

Our mechanistic analysis reveals that Engram effectively “deepen” the network by relieving early layers from static reconstruction tasks, thereby freeing up attention capacity to focus

将外部存储器视为一个可编辑和可扩展的键值存储，使模型能够适应不断变化的信息而无需重新训练。

Mechanisms of Knowledge Storage. 与容量扩展并行，大量研究审视了 Transformer 编码和检索事实知识的内在机制。前馈网络被广泛假设为起到键值记忆的作用 (Geva et al., 2021)。在此框架下，第一层充当模式检测器（”键”），而第二层将特定信息投影到残差流中（”值”）。这种模块化通过识别负责存储不同事实的特定”知识神经元”得到证明 (Dai et al., 2022)。因果追踪方法提供了进一步的验证，该方法将事实回忆的信息流映射到特定的 FFN 层 (Meng et al., 2022)。这些见解催生了精确的模型编辑算法，如 ROME (Meng et al., 2022) 和 MEMIT (Meng et al., 2023)，它们允许直接更新事实关联而无需重新训练。此外，对内部表示的研究，例如在 Othello-GPT (Li et al., 2023a) 中的研究，表明这些存储机制可能促进了结构化”世界模型”的出现，而不仅仅是统计记忆。

7. Conclusion

在这项工作中，我们引入 **conditional memory** 作为主流条件计算范式（MoE）的补充稀疏轴，旨在解决通过动态计算模拟知识检索的低效问题。我们通过 Engram 这一模块实例化了这一概念，该模块将经典的 N-gram 嵌入现代化，以实现针对静态模式的可扩展、恒定时间 $O(1)$ 查找。

通过形式化 *Sparsity Allocation* 问题，我们发现了一种 U 形缩放定律，表明在 MoE 专家和 Engram 记忆之间进行稀疏容量的混合分配，其性能严格优于纯 MoE 基线。在此定律的指导下，我们将 Engram 扩展到 270 亿参数，在多个领域实现了卓越的性能。值得注意的是，虽然记忆模块直观上有助于知识检索，但我们在通用推理、代码和数学方面观察到了更大的收益。

我们的机制分析表明，Engram 通过将早期层从静态重建任务中解放出来，有效地“加深”了网络，从而释放了注意力容量以专注于全局上下文和复杂推理。这种架构转变转化为长上下文能力的显著提升，这在 LongPPL 和 RULER 的性能增益中得到了证明。最后，Engram 倡导将基础设施感知的效率作为首要设计原则。其确定性寻址允许存储与计算的解耦，使得海量参数表可以卸载到主机内存中，而推理开销可忽略不计。我们设想条件记忆功能将成为下一代稀疏模型不可或缺的建模原语。

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on global context and complex reasoning. This architectural shift translates into substantial improvements in long-context capabilities, as evidenced by performance gains in LongPPL and RULER. Finally, Engram advocates for infrastructure-aware efficiency as a first-class design principle. Its deterministic addressing allows for the decoupling of storage and compute, enabling the offloading of massive parameter tables to host memory with negligible inference overhead. We envision conditional memory functions as an indispensable modeling primitive for next-generation sparse models.

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Appendices

A. Detailed Model Architecture and Hyper Parameters

	Dense-4B	**MoE-27B**	**Engram-27B**	**Engram-40B**
Total Params	4.1B	26.7B	26.7B	39.5B
Active Params	3.8B			
Total Tokens	262B			
Layers	30			
Dimension	2560			
Leading Dense Layers	-	1	1	1
Routed Experts	-	72	55	55
Active Experts	-	6	6	6
Shared Experts	-	2	2	2
Load Balancing Method	-	Loss Free (Wang et al., 2024a)		
Attention module	MLA (DeepSeek-AI et al., 2024)			
RoPE θ	10000			
mHC Expansion Rate	4			
Sequence Length	4096			
Vocab Size	129280			
Batch Size	1280			
Training Steps	50000			
Backbone Optimizer	Muon (Jordan et al., 2024)			
Embedding Optimizer	Adam (Kingma, 2014)			
Base Learning Rate	4e-4			
Lr Scheduler	Step Decay (Bi et al., 2024)			
Weight Decay	0.1			
Engram Dim d_{mem}	-	-	1280	1280
Engram Vocab Size	-	-	2262400	7239680
Engram Num Head	-	-	8	8
Engram Layer	-	-	[2,15]	[2,15]
Engram N-gram	-	-	[2,3]	[2,3]
Engram combine mHC	-	-	True	True
Engram tokenizer compression	-	-	True	True
Engram Conv Zero Init	-	-	True	True
Engram Lr Multipler	-	-	x5	x5
Engram Weight Decay	-	-	0.0	0.0
Engram Optimizer (Embed. only)	-	-	Adam (Kingma, 2014)	

Table 5 | Detailed model architecture information and training hyper parameters.

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B. Full Benchmark Curves

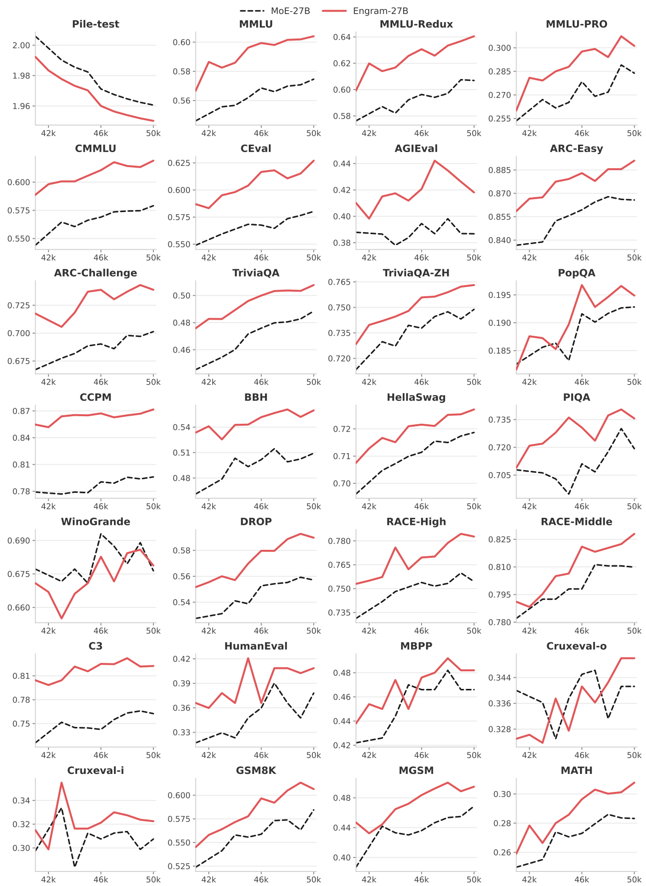

Figure 8 | Last 10k pre-training benchmark curve.

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C. Case Study of Tokenizer Compression

Rank	Merge Count	Normalized Token	Original Tokens
1	163	'␣'	'\t', '\n', '\r', '␣', '␣␣', '\n\n', '␣␣␣', '␣\n', ...
2	54	'a'	'A', 'a', '␣a', '␣A', 'á', 'ä', 'ã', 'ą', '␣à', '␣å', 'â', ...
3	40	'o'	'O', 'o', '␣o', '␣O', 'ó', 'ö', 'ô', 'õ', 'ő', 'ò', ...
4	35	'e'	'E', 'e', '␣e', '␣E', 'é', 'è', '␣é', 'ę', 'ě', 'ê', ...
5	30	'i'	'I', 'i', '␣I', '␣i', 'í', 'ì', 'î', 'ī', 'ï', ...

Table 6 | The table illustrates Top-5 merged tokens by *Tokenizer Compression* and the overall compression ratio is 23.43% for our 128k tokenizer.

Appendices

A. Detailed Model Architecture and Hyper Parameters

B. Full Benchmark Curves

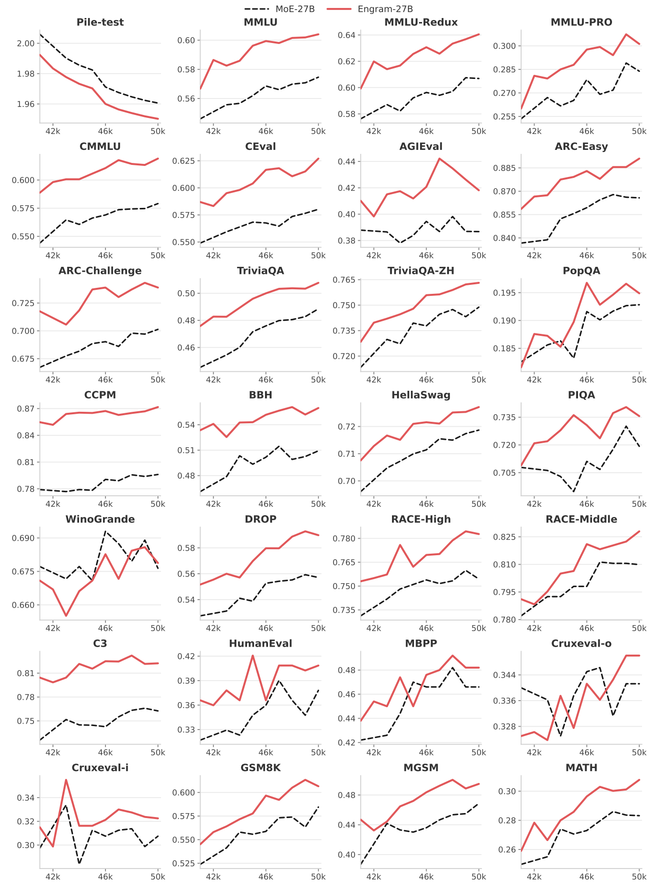

Figure 7 | 最后 10k 步预训练基准曲线。

	Dense-4B	**MoE-27B**	**Engram-27B**	**Engram-40B**
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Engram N-gram	-	-	[2,3]	[2,3]
Engram combine mHC	-	-	True	True
Engram tokenizer compression	-	-	True	True
Engram Conv Zero Init	-	-	True	True
Engram Lr Multipler	-	-	x5	x5
Engram Weight Decay	-	-	0.0	0.0
Engram Optimizer (Embed. only)	-	-	Adam (Kingma, 2014)	

Table 5 | Detailed model architecture information and training hyper parameters.

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Rank	Merge Count	Normalized Token	Original Tokens
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2	54	'a'	'A', 'a', '␣a', '␣A', 'á', 'ä', 'ã', 'ą', '␣à', '␣å', 'â', ...
3	40	'o'	'O', 'o', '␣o', '␣O', 'ó', 'ö', 'ô', 'õ', 'ő', 'ò', ...